ASYMPTOTIC FREQUENCY OF QUADRATIC FIELDS ADMITTING ELLIPTIC CURVES WITH RATIONAL j-INVARIANT AND GOOD REDUCTION EVERYWHERE

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Abstract. We show under the *abc*-conjecture that the number of real and imaginary quadratic fields with rational j-invariant and with discriminant of absolute value at most x with good reduction everywhere is $\sim cx\log^{-1/2}(x)$ and give a formula for c in the case of real and imaginary quadratic fields. A key technical ingredient is progress towards a conjecture of Granville [6]: given a fixed elliptic curve $E/\mathbb{Q}$ with short Weierstrass equation $y^2 = f(x)$ for $f \in \mathbb{Q}[x]$ reducible and without repeated roots, we show that the number of d with $|d| \leq D$ and such that the quadratic twist $dy^2 = f(x)$ has a nontrivial (not 2-torsion) integral point is $\ll D^{2/3+o(1)}$ assuming the *abc*-conjecture. As another intermediate step, we show (unconditionally) that for a polynomially sparse subset S of the natural numbers and any x, the number of pairs of elements that at both most x and which have least-common-multiple at most x is $\ll x^{1-\epsilon}$ for some $\epsilon > 0$.

1. Introduction

An elliptic curve defined over a number field K is said to have "good reduction everywhere" if it has good reduction at every prime ideal of the ring of integers of K. Tate showed that there are no such elliptic curves over $\mathbb{Q}$ [16] but this is not the case for all fields. For example, over $K := \mathbb{Q}(\sqrt{29})$ the elliptic curve

$$y^2 + xy + \left(\frac{5+\sqrt{29}}{2}\right)^2 y = x^3 \tag{1}$$

has good reduction everywhere. Number fields admitting such elliptic curves have been studied extensively, especially over quadratic fields [2, 10–12] but also over higher degree fields [19, 20]. Setzer [18] gave an explicit criterion for m such that $\mathbb{Q}(\sqrt{m})$ admits an elliptic curve with rational j-invariant and good reduction everywhere. We will be interested in the following:

Question. *How often does a real (resp. imaginary) quadratic field admit an elliptic curve with everywhere good reduction and rational j-invariant?*

As with any such question, we must define "often" with respect to an ordering of quadratic fields. Let $R(x)$ (resp. $I(x)$) be the number of real (resp. imaginary) quadratic fields $K/\mathbb{Q}$ with discriminants of absolute value at most x and over which there exist elliptic curves with rational j-invariant and with good reduction at every prime of K. Coupling the criterion of [18] with a lower bound of Serre [17] on the sizes of particular sifted sets of integers, Clemm and Trebat-Leder [1] gave a lower bound on the quantities of interest.

Theorem 1.1 ([1])**.**

$$R(x) \gg \frac{x}{\sqrt{\log x}} \quad \textit{and} \quad I(x) \gg \frac{x}{\sqrt{\log x}}.$$

1.1. **Main Results.** Our main theorem is a sharp result for the asymptotic behavior of $R(x)$ and $I(x)$ assuming the *abc*-conjecture. Because the expressions for the constants are somewhat technical, we defer some of their description to later in the paper.

Theorem 1.2.

$$R(x) \sim \frac{c_R x}{\sqrt{\log x}} \quad \textit{and} \quad I(x) \sim \frac{c_I x}{\sqrt{\log x}}$$

where

$$c_R = \sum_{\substack{d\in\mathbb{Z} \\ d \text{ good}}} \frac{c'_d c''_{d,R}}{d 2^{\omega(d)}} \quad \textit{and} \quad c_I = \sum_{\substack{d\in\mathbb{Z} \\ d \text{ good}}} \frac{c'_d c''_{d,I}}{d 2^{\omega(d)}},$$

where the set of good d is defined in Definition 2.8, c'_d is as in Corollary 5.3 and $c''_{d,R}$ and $c''_{d,I}$ are as in Lemma 5.4.

We may worry that the condition of "rational j-invariant" is too restrictive. However — based on conditionally complete tables of elliptic curves with bounded discriminant and good reduction everywhere due to the first author[1] — most quadratic fields admitting any curves with good reduction everywhere appear to admit at least one with rational j-invariant. We discuss this further in Section 1.3.

An important technical ingredient in the proof of Theorem 1.2 is an upper bound on how often quadratic twists of some elliptic curves over $\mathbb{Q}$ have integral points. Consider an elliptic curve $E/\mathbb{Q}$ with short Weierstrass equation $y^2 = f(x)$. We are interested in counting the number of quadratic twists E_d of E — with models $dy^2 = f(x)$ — that have integral points. Note that if E has a two-torsion point, say $(a : 0 : 1)$, then every twist E_d will also have the two-torsion point $(a : 0 : 1)$. As such, we restrict our attention to determining the existence of a nontrivial integral point — an integral point that is not two-torsion.

Formally, we prove an upper bound on the quantity $|T_E(D)|$, where

$$T_E(D) := \{d \in \mathbb{Z} \colon |d| \le D, d \text{ is squarefree and } E_d \text{ has a nontrivial integral point}\}.$$

In [6], Granville shows the following conditional upper bound on the analogous quantity for hyperelliptic curves.

Theorem 1.3 ([6])**.** *Assume that the abc-conjecture is true. Let C be a hyperelliptic curve given by the integral model $y^2 = f(x)$ where $f \in \mathbb{Z}[x]$ has degree at least three (i.e. the genus is at least one) and does not have repeated roots. Then,*

$$|T_C(D)| \ll D^{\frac{1}{\deg f - 2} + o(1)}.$$

Unfortunately, in the case of elliptic curves, $\deg f = 3$ so Theorem 1.3 does not give us any information. However, Granville also conjectures that the $\deg f - 2$ in the denominator of the exponent can be replaced with $\deg f$.

Conjecture 1.4 ([6])**.** *Let C be a hyperelliptic curve given by the integral model $y^2 = f(x)$ where $f \in \mathbb{Z}[x]$ has degree at least three (i.e. the genus is at least one) and does not have repeated roots. Then,*

$$|T_C(D)| \sim \kappa_f D^{\frac{1}{\deg f} + o(1)},$$

where κ_f is some constant that can be determined explicitly given f.

Granville proves Conjecture 1.4 for f with degree at least 7 which split into linear factors over $\mathbb{Q}$. We make progress towards Conjecture 1.4 in the setting of elliptic curves.

Theorem 1.5. *Assume that the abc-conjecture is true. Let E be an elliptic curve over $\mathbb{Q}$ with short Weierstrass equation $y^2 = f(x)$ where $f(x) \in \mathbb{Z}[x]$ is reducible (over $\mathbb{Q}$) but has no repeated roots (over $\mathbb{C}$). Then, for any $\epsilon > 0$*

$$|T_E(D)| \ll_\epsilon D^{\frac{2}{3}+\epsilon}.$$

[1]The tables are complete assuming the generalized Riemann hypothesis.

We may apply Theorem 1.5 to the set of good d in Theorem 1.2 to show (among other things) that the sum converges. At a high level, this sum is actually a union bound over the good $d \subseteq \mathbb{Z}$, and thus yields an upper bound on the leading constant in Theorem 1.2. To show the matching lower bound, we control the second term in the inclusion-exclusion sequence via a result on the sizes of pairwise least-common-multiples of polynomially sparse subsets. This result may be of independent interest.

Theorem 1.6. *Given a set $S \subseteq \mathbb{N}$ of squarefree numbers and positive constants B and $\beta < 1$ such that*

$$\#\{n \leq x \colon n \in S\} \leq Bx^{1-\beta}$$

for all positive x, there are positive constants C and $\kappa < 1$ such that

$$\#\{(n, n') \colon n \in S, n' \in S, \operatorname{lcm}(n, n') \leq x\} \leq Kx^{1-\kappa}.$$

for all positive x.

Corollary 1.7. *The set of pairwise least-common-multiples of a polynomially sparse set of squarefree numbers is polynomially sparse.*

1.2. **Techniques.** The first input used in proving Theorem 1.2 is the aforementioned criterion of Setzer [18] for when $\mathbb{Q}(\sqrt{m})$ admits an elliptic curve with good reduction everywhere and rational j-invariant. We state this criterion fully as Theorem 2.10 in Section 2 but for now will give the following high-level overview. Theorem 2.10 tells us that $\mathbb{Q}(\sqrt{m})$ admits such an elliptic curve if m can be factored as nd for some d such that

(a) d is the squarefree part of $a^3 - 1728$ for a in some positive density subset of the integers.
(b) n is divisible only by primes satisfying certain quadratic residuosity conditions with respect to d. For any d, the set of primes satisfying this condition has natural density $\frac{1}{2}$;
(c) The image of n in $(\mathbb{Z}/4d\mathbb{Z})^\times$ must lie in a specified subset.

We will describe our methods for $R(x)$ but the same ideas are used for $I(x)$. We first study the asymptotics of the quantity $R_d(\frac{x}{|d|})$, which we can think of the contribution to $R(x)$ of those m which are divisible by d. We then aggregate over all d satisfying (a) above to obtain bounds on $R(x)$. Our approach was motivated by the lower bound of Clemm and Trebat-Leder (Theorem 1.1). They start by choosing a single value of d satisfying (a) and for which (c) is trivial. They then use a result of [17] giving the size of sifted sets of integers which applies when the sieving set is primes satisfying residuosity conditions. This allows them to lower bound the number of n satisfying (b) and show that for this choice of d, $R_d(\frac{x}{|d|}) \gg \frac{x}{\sqrt{\log x}}$. This may seem surprising, as it means that even without considering multiple values of d they are already able to obtain the correct order of growth! This happens because the number of d satisfying (a) is very sparse, as we prove in Theorem 1.5 assuming the *abc*-conjecture. Therefore, aggregating the contributions from all d satisfying (a) does not change the overall answer by more than a constant. More formally, we have

$$R(x) \leq \sum_{d \text{ sat. (a)}} R_d\left(\frac{x}{|d|}\right). \tag{2}$$

The first step in our proof is showing the upper bound

$$R_d(y) \ll \frac{(1 + o_d(a))y}{\sqrt{\log y}}. \tag{3}$$

The key input for (3) is Selberg-Delange theory (in particular, Theorem 2.2), which is also the general theory underlying the bound of [17] used in [1]. Selberg-Delange theory gives us a Tauberian theorem for Dirichlet series which can be expressed as $\zeta(s)^\rho G(s)$ for $\rho > 0$ and $G(s)$ holomorphic in a neighborhood around $s = 1$. It tells that the sum of coefficients of the series up to y is $\sim \frac{cy}{\log^{1-\rho} y}$ for some explicit constant c depending on F. To obtain our upper bound, we apply Selberg-Delange

theory to to the Dirichlet series $F(s)$ of the characteristic function of (b)[2]. It turns out that because the number of "valid" primes in (b) is half of all primes, $F(s) = \zeta(s)^{\frac{1}{2}} G(s)$ for some $G(s)$ holomorphic around $s = 1$. This then implies (3).

Applying (3) to (2), we have

$$R(x) \ll \sum_{d \text{ sat. (a)}} \frac{(1 + o_d(a))x}{|d|\sqrt{\log x - \log |d|}}.$$

Summation by parts shows that the sum of the reciprocals of the elements of a polynomially sparse set is constant (Lemma 2.1), and applying this to the set of d satisfying (a) (which is polynomially sparse by Theorem 1.5), we have that the series

$$\sum_{d \text{ sat. (a)}} \frac{1}{|d|}$$

converges. This allows us to let d go to infinity slowly with respect to x. In this regime, we can show that the $o_d(a)$ terms are uniformly bounded and obtain

$$R(x) \ll \frac{x}{\sqrt{\log x}} \sum_{d \text{ sat. (a)}} \frac{1}{|d|} \ll \frac{x}{\sqrt{\log x}}.$$

In order to obtain the precise constants, we start by computing $R_d(y)$ exactly (Lemma 5.4). To do this, we need to fix a d and count those n which satisfy (b) and (c) above. As mentioned above, we can use Selberg-Delange theory to obtain the exact constant if we are only interested in the coefficients of the Dirichlet series of (b). Observe that the Dirichlet series of (b)$\wedge$(c)[3] is the (Rankin-Selberg) convolution of the Dirichlet series for (b) and the Dirichlet series for (c). The tricky part is incorporating (c), which in general need not be a multiplicative property — because of this, the Dirichlet series of (c) need not have an Euler product. We deal with this issue by expression (c) as a linear combination of Dirichlet L-series. Then, the convolution of the Dirichlet series for (b) and the Dirichlet series for (c) is a linear combination of convolutions of L-functions. Intuitively, a Tauberian theorem can be thought of as telling us the "rate of divergence" at a pole, and hence only terms which possess a pole at $s = 1$ will contribute to the overall asymptotics. By choosing our expansion of (c) into Dirichlet L-series carefully, we make it so that only one term of the linear combination contributes to the final asymptotics and apply Theorem 2.2 to this term to obtain the final answer. This yields a precise enough answer for $R_d(y)$. The constant ultimately contains an infinite product of form $\prod_{q \in S}(1 - \frac{1}{q^2})$ over the set of primes satisfying (b). Because we can explicitly determine when q is in S, we are able to compute the constant in (3) to arbitrary accuracy.

We are now left with the problem of replacing our upper bound in (2) with a precise statement. Observe that (2) is simply a union bound over the contributions of all d satisfying (a). We apply the principle of inclusion-exclusion and lower bound $R(x)$ by the sum of the first two terms in the inclusion-exclusion series. The first term is simply the union bound that we have already computed. The second term is a sum over pairs (d, d'), both satisfying (a), where each term accounts for the contribution to $R(x)$ of those n which are divisible by $\operatorname{lcm}(|d|, |d'|)$. Thus, abusing notation, we want to upper bound

$$\sum_{d,d' \text{ sat. (a)}} R_{d,d'}(x), \tag{4}$$

where each term captures the relevant contribution from n dividing both d and d'. The precise asymptotics of $R_{d,d'}(y)$ depends on several inputs including most notably the compatibility between

[2]By this, we mean the Dirichlet series where the coefficient of n is 1 if n satisfies (b) and is 0 otherwise.

[3]The Dirichlet series whose coefficient at n is 1 if and only if n satisfies (b) and (c).

d and d'. However, as an upper bound it suffices to count the number of n which satisfy (b) with respect to both d and d'. Intuitively, each d multiplies the natural density of the valid set of primes by a factor of $\frac{1}{2}$ and we are left with those n dividing only primes in a subset containing $1/4$ of all primes. This tells us that for a fixed (d, d'), $R_{d,d'}(y)$ can be upper bounded as

$$R_{d,d'}(y) \asymp \frac{y}{\operatorname{lcm}(|d|, |d'|) \log^{3/4} y}. \tag{5}$$

We want to repeat the argument we used to upper bound (2) to pass from (5) to an upper bound on (4). This means that we need to show that

$$\sum_{d,d' \text{ sat. (a)}} \frac{1}{\operatorname{lcm}(|d|, |d'|)}$$

converges. We do so by proving Theorem 1.6, which tells us that the number of pairs of elements up to x in a polynomially sparse set with lcm at most x is $\ll x^{1-\kappa}$ for some $\kappa > 0$. We can then use summation by parts as before to show that the sum in question converges to a constant independent of x. This means that the second term of the inclusion-exclusion series is $\ll \frac{x}{\log^{3/4} x}$ and is asymptotically negligible compared to the first term. Therefore, the constant we obtained from the union bound is actually the correct constant.

1.2.1. *Twists of elliptic curves: proving Theorem 1.5.* The key idea motivating the application and proof of Theorem 1.5 is the relationship between squarefree parts and quadratic twists of elliptic curves. Consider (a) above. We have that d is the squarefree part of $r^3 - 1728$ if and only if for some integer t we have $dt^2 = r^3 - 1728$. This happens if and only if the dth quadratic twist of $r^3 - 1728$ has an integral point. We are not interested in solutions where $t = 0$ and $r = 12$ because these exist for every d — these solutions correspond to the 2-torsion point $(12 : 0 : 1)$ of $dy^2 = x^3 - 1728 = (x - 12)(x^2 + x + 144)$. By definition $T_E(D)$ counts the nontrivial points and hence the number of d which arise as squarefree parts of $x^3 - 1728$.

The proof of Theorem 1.5 has two main cases — the others reduce to these two via coordinate transformations. The first case, when f factors completely into linear polynomials, follows immediately from an argument in Granville's proof of Theorem 1.3. The second case, when f factors as $(x - a)(x^2 - b)$ for b not a square, is our contribution. It is also the one that applies to our curve of interest, $y^2 = x^3 - 1728$. We want to count the number of d with $|d| \le D$ such that d arises as the squarefree part of $a^3 - 1728$. We first use a result of Granville (Theorem 2.6) that, assuming the *abc*-conjecture, tells us that these small d do not arise as the squarefree parts coming from a which are large in absolute value. This gives us a bound on which a we need to consider. We also have (Lemma 3.2) that the squarefree part of $a^3 - 1728$ is the product of the squarefree parts of $a - 12$ and $a^2 + 12a + 144$ away from finitely many primes, enabling us to treat the two factors separately. The key idea ends up being that for any squarefree number d_2, there are not many values of a such that the squarefree part of $a^2 + 12a + 144$ is d_2. This is because, rewriting $a^2 + 12a + 144$ as $b^2 + 108$ by a change of variables, if $b^2 + 108 = d_2 t^2$ and $b'^2 + 108 = d_2 t'^2$, b and b' then $b + t\sqrt{d_2}$ and $b' + t'\sqrt{d_2}$ have the same norm in $\mathbb{Q}(\sqrt{d_2})$. This means that (up to sign) they differ by a power of a fundamental unit, and we use this to show that there are at most $\ll \log D$ different values of b (and hence a) which yield this value of d_2. After some casework, this yields our result.

Note that all we need in order to show that the series

$$\sum_{d \text{ sat. (a)}} \frac{1}{|d|}$$

converges is that the set of d which satisfy (a) and have $|d| \le D$ has size that is $\ll D^{1-\kappa}$. Because we are only interested in convergence of this series, we do not need an explicit form of the *abc*-conjecture in order to reach this conclusion.

1.2.2. *Least-common-multiples of polynomially sparse sets: proving Theorem 1.6.* This proof is motivated by the analogous result when least-common-multiple is replaced with product. In this case, consider the number of tuples $(a, b) \in S \times S$ where $y \leq a < 2y$, $b \leq x$, and $ab \leq x$. There are $\ll y^{1-\epsilon}$ choices for a and, for each, $\ll (\frac{x}{y})^{1-\epsilon}$ choices for b. Therefore, there are $\ll x^{1-\epsilon}$ tuples with a in $[y, 2y)$. Partitioning the range $[1, x]$ into such intervals and summing over the $\lceil \log x \rceil$ choices of y we see that there are the total number of tuples is $\ll x^{1-\epsilon} \log x$.

In the least-common-multiple (henceforth LCM) setting, we start by restricting to counting pairs $(a, b) \in S \times S$ where $y \leq a < 2y$ and $z \leq b < 2z$. The primary obstruction to mimicking the approach we used for products to LCMs is that the LCM of a and b is $\frac{ab}{(a,b)}$, which will in general be smaller than ab. In the product case, we multiplied together the number of possible values of a and the number of possible values of b for each a. In the LCM case, we will multiply together the number of possible values of a, the number of factors of a which are at least $\frac{az}{x}$ (which we think of as the number of possible greatest-common-divisors of a and something else), and the number of values of b which are divisible by the chosen factor of a. Note that the greatest-common-divisor of a and b must be at least $\frac{az}{x}$ because otherwise $\mathrm{lcm}(ab) = \frac{ab}{(a,b)} > \frac{bx}{z} \geq x$. Counting the number of possible values of a and counting the number of possible values of b which are divisible by the chosen factor of a ends up being straightforward. The remaining term in the product comes from the number of large factors of a — those which are at least $\frac{az}{x}$. These are in bijection with small factors of a — those which are at most $\frac{x}{z}$. If z is large we bound this number simply by $\frac{x}{z}$. If z is small we note that we are interested in the number of numbers up to $\frac{x}{z}$ which are divisible only by the $\leq \lceil \log y \rceil + 1$ prime factors of a. By an upper bound on smooth numbers based on Rankin's trick [7], we can show that this count is small. We then obtain the final number of tuples by multiplying by $\lceil \log x \rceil^2$ to account for the choices of y and z.

1.3. **Future Work.** There are several natural extensions. The first concerns the generalization of our result to elliptic curves with good reduction everywhere and any j-invariant.

Conjecture 1.8. *Theorem 1.2 holds even after removing the constraint that j is rational.*

To show this, we would want to show that the number of real and imaginary quadratic fields with discriminant of absolute value at most x and over which there exists an elliptic curve with good reduction everywhere but no elliptic curve with good reduction everywhere and rational j-invariant is $o(\frac{x}{\sqrt{\log x}})$. As mentioned, this conjecture is motivated by elliptic curve tables constructed by the first author [15] — of the 955 quadratic fields admitting an elliptic curve with good reduction everywhere, only 130 of the fields do not admit such a curve that also has rational j-invariant. While there is no criterion as explicit as that of Setzer [18] that detects elliptic curves with irrational j-invariant, Cremona and Lingham [3] give an algorithm for computing such fields along with some criteria in terms of the existence of integral solutions to certain Diophantine problems over $\mathcal{O}_K$. In the rational case, we used the *abc*-conjecture to bound the number of solutions to the analogous Diophantine problems (over $\mathbb{Z}$), so perhaps an approach based on generalizations of the *abc*-conjecture to number fields holds promise.

The dependence on the *abc*-conjecture arises whenever we use Theorem 2.6 to bound the range of a for which it is possible for the squarefree part of $a^3 - 1728$ to be d. There are unconditional results for the density of inputs on which a cubic polynomial is squarefree [9], and techniques for studying squarefree parts of polynomials often go through techniques for studying squarefree values of polynomials. As such, it is plausible that these methods could be used to remove this dependence.

1.4. **Organization.** Most preliminary content, including proofs of elementary results and references to the literature, are in Section 2. We prove our bound on how often twists of some elliptic curves have an integral point, Theorem 1.5, in Section 3. In Section 4, we apply Selberg-Delange theory and Theorem 1.5 to prove Theorem 4.1, a version of Theorem 1.2 that is tight up to constants.

In Section 5, we compute an upper bound on the leading constants in Theorem 1.2. In Section 6, we prove Theorem 1.6 and use it to prove a matching lower bound on the leading constants of Theorem 1.2, concluding the proof.

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2. Preliminaries

2.1. **Notation.** We write $\mathbb{N}$ to denote the positive integers. In general, p and q will be used to denote primes and $\prod_p$, $\prod_q$, $\sum_p$, and $\sum_q$ denote products and sums over primes. For p a prime, we write $|\cdot|_p$ to denote the p-adic norm. Given a number field $K/\mathbb{Q}$, we write Δ_K to denote its absolute discriminant.

Let n be an integer. We write $\omega(n)$ to denote the number of distinct prime factors of n. Generally, we will apply this in contexts where n is squarefree in which case $\omega(n)$ is simply the number of factors of n. If t is the largest integer for which t^2 divides n then we call $d := \frac{n}{t^2}$ the **squarefree part** of n and write $d = \mathrm{sqf}(n)$. The product of the distinct prime factors of n is the radical of n, which we denote by $\mathrm{rad}(n)$. Note that the squarefree part of n includes its sign but the radical does not. Given two positive integers m and n, we write (m, n) to denote their greatest-common-divisor.

Throughout, we use Vinogradov asymptotic notation. If $f \ll g$ (resp. $f \gg g$) if $\limsup_{x\to\infty} \frac{|f(x)|}{g(x)}$ (resp. $\limsup_{x\to\infty} \frac{|g(x)|}{f(x)}$). If $f \ll g$ and $g \ll f$ then we write $f \asymp g$. We will also occasionally make use of Knuth asymptotic notation to concisely describe error. The expression $b(x) = c(x) + o(g(x))$ means that $\limsup_{x\to\infty} \frac{b(x)-c(x)}{|g(x)|} = 0$. Similarly, we write $b(x) = c(x) + O(g(x))$ when $b(x) - c(x) \ll g(x)$. A subscript on any such notation — for example, $\gg_\epsilon$ — means that the implicit function or constant may depend on the subscript.

Given complex numbers z and a, we define the power $z^a := e^{a \log z}$ with respect to the principal branch of the logarithm.

We will often be concerned with Dirichlet characters of modulus 8. To this end, it will help to fix a notation for characters of $(\mathbb{Z}/8\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$. Without loss of generality, let 3 (mod 8) correspond to $(1, 0) \in \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, 5 (mod 8) to $(0, 1)$, and 7 (mod 8) to $(1, 1)$. We then define the characters χ_{ij}, for $i, j \in \{0, 1\}$, to be nontrivial on the first (resp. second) component when i (resp. j) is 1.

2.2. **Analytic Facts.** The following lemma will allow us to convert results on the sparseness of a subset of the natural numbers to results on the sums of reciprocals of elements of that subset.

Lemma 2.1. *Let $\alpha < 1$ and C be positive constants. Suppose that a multiset of the natural numbers $f\colon \mathbb{N} \to \mathbb{N}$ is such that for all x,*

$$\sum_{n=1}^{x} f(n) \le Cx^\alpha.$$

Then,

$$\sum_{n=1}^{\infty} \frac{f(n)}{n} \le C\zeta(2-\alpha),$$

and in general for $m \ge 2$

$$\sum_{n=m}^{\infty} \frac{f(n)}{n} \le \frac{2}{1-\alpha} m^{\alpha-1},$$

Proof. Let $\pi_\Lambda(y) := \sum_{n=1}^{y} f(n)$. We apply summation by parts.

$$\sum_{n=m}^{x} \frac{f(n)}{n} = \sum_{n=m}^{x} \frac{1}{n}(\pi_\Lambda(n) - \pi_\Lambda(n-1))$$
$$= \frac{\pi_\Lambda(x)}{x} - \frac{\pi_\Lambda(m-1)}{1} + \sum_{n=m}^{x-1} \left(\frac{1}{n} - \frac{1}{n+1}\right) \pi_\Lambda(n).$$

In the limit as x goes to infinity, we have

$$\leq C \sum_{n=m}^{\infty} \frac{1}{n^{2-\alpha}}$$

and the two parts of the lemma follow. □

We will also use theorem from Selberg-Delange theory (in turn based on a Tauberian theorem of Ikehara) to obtain asymptotic estimates for sums of multiplicative functions from properties of their Dirichlet series.

Theorem 2.2 (Special case of Chapter 5 Theorem 3 from [21])**.** *Suppose that $F(s) := \sum_n b_n n^{-s}$ is a Dirichlet series such that $b_n \geq 0$ for all n. Suppose further that for some $\rho \neq 0$, $G(s) := F(s)\zeta(s)^\rho$ can be analytically continued to a holomorphic function on the region $\mathrm{Re}(s) \geq 1-\beta$ for some $\beta > 0$, and in this region satisfies the bound*

$$|G(s)| \leq M(1 + |\mathrm{Im}(s)|)^{1-\delta} \tag{6}$$

for some $M > 0$ and $0 < \delta \leq 1$. Then,

$$\sum_{n \leq x} b_n = \frac{x}{\log^{1-\rho} x} \left(\frac{G(1)}{\Gamma(\rho)} + O(Me^{-K\sqrt{\log x}}) \right)$$

where K and the implicit constant are absolute.

Theorem 2.3 ([13])**.** *Let χ be a nonprincipal Dirichlet character with modulus m. Let $\frac{1}{2} \leq \mathrm{Re}(s) \leq 1$ and $w := \frac{\log m}{\log\log m}$. Then,*

$$|L(s,\chi)| \ll (2 + \mathrm{Im}(s))^{-\mathrm{Re}(s)} m^{\frac{3}{8}(1-\mathrm{Re}(s))} C^w \log(2 + \mathrm{Im}(s)).$$

where C and the implicit constant are absolute.

Proposition 2.4 (Subsection 3.10 of [7])**.** *Let $\Psi(u,v)$ denote the number of positive integers up to u which are divisible only by primes up to v. Then, if $A > 1$, then*

$$\Psi(u, \log^A u) = u^{1-\frac{1}{A}+O(\frac{1}{\log\log u})}. \tag{7}$$

2.3. **The *abc*-conjecture and some consequences.** Recall the *abc*-conjecture.

Conjecture 2.5 (Oesterlé, Masser, Szpiro)**.** *For every $\epsilon > 0$ there exists a constant C_ϵ such that, given coprime positive integers a, b, and c such that for some selection of signs, $\pm a \pm b \pm c = 0$.*

$$\max(|a|, |b|, |c|) \leq C_\epsilon \,\mathrm{rad}(abc)^{1+\epsilon}$$

Theorem 2.6 (Corollary 1 of [5])**.** *Assume that the abc-conjecture (Conjecture 2.5) is true. Suppose that $g(x) \in \mathbb{Z}[x]$ has no repeated roots. Then, for any $r \in \mathbb{Z}$,*

$$\mathrm{rad}\, g(r) \gg_\epsilon |r|^{\deg g - 1 - \epsilon}.$$

For a hyperelliptic curve with integral model $C\colon y^2 = f(x)$, we write C_d to denote its dth quadratic twist, which has model $dy^2 = f(x)$. We will make essential use of the following theorem.

Theorem 2.7 (Theorem 1(i) of [6]). *Assume that the abc-conjecture is true. Suppose that $f(x) \in \mathbb{Z}[x]$ does not have repeated roots and let C be the hyperelliptic curve with equation $y^2 = f(x)$. If $\deg f \geq 3$ then the integral points on C_d with x-coordinate r satisfy*

$$|r| \ll_\epsilon |d|^{\frac{1}{\deg f - 2} + \epsilon}$$

and

$$|t| \ll_\epsilon |d|^{\frac{1}{\deg f - 2} + \epsilon}$$

Proof. Let $dt^2 = f(r)$. It follows from Theorem 2.6 that, under the *abc*-conjecture

$$|d|^{\frac{1}{2}} |r|^{\frac{\deg f}{2}} \gg_g |df(r)|^{\frac{1}{2}} = |dt| \geq \operatorname{rad}(dt) = \operatorname{rad}(f(r)) \gg_\epsilon |r|^{\deg f - 1 - \epsilon}$$

and the first part of the result follows.

For the second part, note that

$$|dt^2| = |f(r)| \ll_g |r|^{\deg f} \ll |d|^{\frac{\deg f}{\deg f - 2} + \epsilon}.$$

□

2.4. **Identifying quadratic fields with good reduction everywhere.** The following definition is the formal statement of condition (a) from Section 1.2.

Definition 2.8. We say that $d \in \mathbb{Z}$ is **good** if $d = \operatorname{sqf}(r^3 - 1728)$ for r a member of the set

$$\{n \in \mathbb{Z} \colon n \equiv 0 \pmod 2 \implies n \equiv 0, 4 \pmod{16}, n \equiv 0 \pmod 3 \implies n \equiv 12 \pmod{27}\}. \tag{8}$$

Definition 2.9. Given $d \in \mathbb{Z}$ squarefree,

$$\epsilon_d := \begin{cases} 1 & d \equiv 1 \pmod 4 \\ -1 & \text{otherwise} \end{cases}$$

We may now state a criterion of Setzer [18] which tells us when a quadratic field $\mathbb{Q}(\sqrt{m})$ admits an elliptic curve with good reduction everywhere and rational j-invariant. This will formalize conditions (b) and (c) from Section 1.2.

Theorem 2.10 (Theorem 2.2 of [1], correcting an error in Theorem 2 of [18]). *Fix d good, and $m = nd$ square-free. $K = \mathbb{Q}(\sqrt{m})$ admits a curve with good reduction everywhere if and only if the following conditions are satisfied.*

(1) $\left(\frac{\epsilon_d d}{q_j}\right) = 1$ *for all primes q_j dividing n;*
(2) $\left(\frac{-\epsilon_d n}{p_i}\right) = 1$ *for all primes p_i dividing d;*
(3) If $d \equiv \pm 3 \pmod 8$ then $n \equiv d \pmod 4$;
(4) If d is even then $n \equiv d + 1 \pmod 8$.
(5) $m > 0$ if $\epsilon_d d < 0$;

In the above, neither d nor n are restricted to being positive integers. Intuitively, the first condition of Theorem 2.10 is (b) from Section 1.2, and the second, fourth, and fifth conditions of Theorem 2.10 is (c) from Section 1.2.

We will now define the primary quantities of interest in this paper. We write that an (elliptic) curve over a field $K/\mathbb{Q}$ has $\mathrm{GRE}_\mathbb{Q}$ if it has good reduction at every prime and j-invariant in $\mathbb{Q}$.

Definition 2.11. Following [1], we define for every positive x

$$R(x) := \#\{m \colon 0 < \Delta_{\mathbb{Q}(\sqrt{m})} \leq x, \mathbb{Q}(\sqrt{m}) \text{ admits an elliptic curve with } \mathrm{GRE}_\mathbb{Q}\}$$

and

$$I(x) := \#\{m \colon -x \leq \Delta_{\mathbb{Q}(\sqrt{m})} < 0, \mathbb{Q}(\sqrt{m}) \text{ admits an elliptic curve with } \mathrm{GRE}_\mathbb{Q}\}.$$

If d is good per Definition 2.8, we also define

$$R_d(x) := \#\{n\colon 0 < \Delta_{\mathbb{Q}(\sqrt{nd})} \leq x, \mathbb{Q}(\sqrt{nd}) \text{ admits an elliptic curve with } \mathrm{GRE}_{\mathbb{Q}}\},$$

and

$$I_d(x) := \#\{n\colon -x \leq \Delta_{\mathbb{Q}(\sqrt{nd})} < 0, \mathbb{Q}(\sqrt{nd}) \text{ admits an elliptic curve with } \mathrm{GRE}_{\mathbb{Q}}\}.$$

Intuitively, $R_d(x)$ and $I_d(x)$ measure the contribution of m which are divisible by d to $R(x)$ and $I(x)$ respectively. Our definitions of $R_d(x)$ and $I_d(x)$ differ somewhat from those of [1]. They write $R_d(x)$ (and $I_d(x)$ analogously) to count the number of n up to x such that $\mathbb{Q}(\sqrt{nd})$ admits an elliptic curve with $\mathrm{GRE}_{\mathbb{Q}}$.

3. Twists of elliptic curves

We will start by defining the quantity we wish to bound. Throughout this section, we will take E to be an elliptic curve defined over $\mathbb{Q}$ with short Weierstrass equation $y^2 = f(x)$ for $f(x) \in \mathbb{Z}[x]$.

Definition 3.1. Let E be an elliptic curve defined over $\mathbb{Q}$ with short Weierstrass equation $y^2 = f(x)$ for $f(x) \in \mathbb{Z}[x]$. We write E_d, the dth quadratic twist of E, to denote the curve given by the integral model $dy^2 = f(x)$. Then,

$$T_E(D) := \{d \in \mathbb{Z}\colon |d| \leq D, d \text{ is a squarefree integer, and } E_d \text{ has an integral point without 2-torsion}\}.$$

It will be useful to reformulate bounding $T_E(D)$ as bounding the number of integers with absolute value up to D that arise as squarefree parts of $f(x)$. To see why, notice that the equation $dy^2 = f(x)$ has an integral point if and only if d is the squarefree part of $f(r)$ for some integer r. Observe that the squarefree part function is multiplicative but not totally multiplicative. Nonetheless, the following lemma will let us write the squarefree part of a polynomial without repeated roots as the product of the squarefree parts of its factors if we ignore the valuations at finitely many primes.

Lemma 3.2. *Suppose that $f \in \mathbb{Z}[x]$ has no repeated roots over $\mathbb{C}$ and factors as $f = \prod_i g_i$ for $g_i \in \mathbb{Z}[x]$. Then, for all but finitely many p and for all $r \in \mathbb{Z}$,*

$$|\operatorname{sqf}(f(r))|_p = \prod_i |\operatorname{sqf}(g_i(r))|_p$$

Proof. If two distinct factors g_i and g_j share a root modulo a prime then the resultant of $\bar{g}_i$ and $\bar{g}_j$ in $\mathbb{F}_p$ must be 0, meaning that p must divide the resultant of g_i and g_j. By assumption, g_i and g_j have no shared roots over $\mathbb{C}$ and hence their resultant is a nonzero integer. This means that g_i and g_j can only share a root modulo the finitely many primes that divide their resultant. Taking the union of these finite sets over all pairs of distinct i and j, we see that for p outside this finite union, if p divides $g_i(r)$ for some r then p does not divide $g_j(r)$ for all $j \neq i$. This means that the factors are "coprime" outside our finite set of primes, and the lemma then follows from the multiplicativity of the squarefree part function. □

We recall the statement of Theorem 1.5.

Theorem 1.5. *Assume that the abc-conjecture is true. Let E be an elliptic curve over $\mathbb{Q}$ with short Weierstrass equation $y^2 = f(x)$ where $f(x) \in \mathbb{Z}[x]$ is reducible (over $\mathbb{Q}$) but has no repeated roots (over $\mathbb{C}$). Then, for any $\epsilon > 0$*

$$|T_E(D)| \ll_\epsilon D^{\frac{2}{3}+\epsilon}.$$

Proof of Theorem 1.5. There are four cases depending on how $f(x)$ factors. In the definition of each case, each a_i and b_j is an integer and we factor $f(x)$ as must as possible in $\mathbb{Q}$.

(1) $\boldsymbol{f(x) = (x + a_1)(x + a_2)(x + a_3)}$: This case follows directly from the proof of Theorem 2 in [6]. For the reader's convenience, we include a sketch of Granville's proof. To get an upper bound on the number of squarefree d for which there exist (r, t) such that $f(r) = dt^2$, we will upper bound the number of pairs (r, t) such that t^2 divides $f(r)$ and $f(r)/t^2$ is squarefree. We will restrict to counting pairs (r, t) which satisfy the given conditions and for which $R < r \leq 2R$ and $Y < d \leq 2Y$ since summing over all these cases at the end only costs us a factor of $\log^2 D$ by Theorem 2.7. If R is small relative to D (for example, we can take $R \ll D^{4/9}$ in the proof), the range of possible values of t is small. We may then obtain an upper bound by summing over the possible values that t can take and for each, bounding the number of r in the relevant range such that $f(r) \equiv 0 \pmod{t}^2$. This ultimately nets us a bound of $\ll_\epsilon D^{1/3+\epsilon}$ for any $\epsilon > 0$.

When R is somewhat large relative to D — say, $D^{8/21} \ll R \ll_\epsilon D^{1+\epsilon}$ — and t has a large range of possible values, we instead argue that any t in a valid pair must have some divisor τ in the range $R^{1/3} < \tau \leq R^{2/3}$. We can thus sum over τ and bound the number of r in the relevant range such that $f(r) \equiv 0 \pmod{\tau}^2$. In this regime, this is better than the previous argument and we end up with a bound of $\ll R^{2/3} \log^3 3R$. By assumption, this can be made $\ll_\epsilon D^{2/3+\epsilon}$.

If we make the implicit constant above a sufficiently large function of $\frac{1}{\epsilon}$, we do not need to consider larger values of R by Theorem 2.7.

(2) $\boldsymbol{f(x) = (x - a)(x^2 - b)}$: We have

$$\operatorname{sqf}(f(r)) = \frac{1}{c(r)^2} \operatorname{sqf}(r - a) \operatorname{sqf}(r^2 - b),$$

where $c(r)$ corresponds to primes which divide both $r - a$ and $r^2 - b$. By Lemma 3.2, the set S of primes p for which there exists an r such that p divides both $r - a$ and $r^2 - b$ is finite, and therefore

$$\operatorname{sqf}(f(r)) \leq \frac{1}{C^2} \operatorname{sqf}(r - a) \operatorname{sqf}(r^2 - b),$$

where $K := \prod_{p \in S} p$. Choose an $\epsilon > 0$, suppose the constant from Theorem 2.7 for this ϵ is C_ϵ, and let

$$T'_E(D) := \{r \colon |r| \leq C_\epsilon D^{1+\epsilon}, |\operatorname{sqf}(r - a) \operatorname{sqf}(r^2 - b)| \leq K^2 D\}.$$

Notice that $T_E(D)$ contains small values of d while $T'_E(D)$ contains those r for which $\operatorname{sqf}(f(r))$ is small. We claim that $|T_E(D)| \leq |T'_E(D)|$. This is because if $d \in T_E(D)$ is the squarefree part of $f(r)$, then automatically $|\operatorname{sqf}(r - a) \operatorname{sqf}(r^2 - b)| = c(r)^2 d \leq K^2 D$ so $r \in T'_E(D)$. We only need to consider $|r| \leq C_\epsilon D^{1+\epsilon}$ by Theorem 2.7.

We will branch into two subcases. First, consider r such that $|\operatorname{sqf}(r - a)| \leq D^{2\delta}$ for some $\delta > 0$ that we will select later. For each squarefree number $m \leq D^{2\delta}$, there are $\leq 2\sqrt{\frac{C_\epsilon D^{1+\epsilon}}{m}} + 1$ values of r such that $\operatorname{sqf}(r - a) = m$ and $|r| \leq C_\epsilon D^{1+\epsilon}$. Therefore, the contribution to $T'_E(D)$ from these r is at most

$$\begin{aligned} 2\sqrt{C_\epsilon} \int_1^{D^{2\delta}} \left(\frac{D^{1+\epsilon}}{z}\right)^{\frac{1}{2}} dz &= 2C_\epsilon^{\frac{1}{2}} D^{\frac{1}{2}(1+\epsilon)} \int_1^{D^{2\delta}} z^{-\frac{1}{2}} dz \\ &\leq C_1 D^{\frac{1}{2}(1+\epsilon)+\delta}. \end{aligned}$$

for $C_1 \leq 4C_\epsilon^{\frac{1}{2}}$.

Second, we will count r contributing to $T'_E(D)$ for which $|\operatorname{sqf}(r-a)| > D^{2\delta}$. Because we require $|\operatorname{sqf}(f(r))| \leq K^2 D$, we require that $|\operatorname{sqf}(r^2-b)| \leq D^{1-2\delta}$. Fix an integer m such that $|m| \leq D^{1-2\delta}$. We will bound the number of r for which $\operatorname{sqf}(r^2-b) = m$. For any such r, there is an $s \in \mathbb{Z}$ such that

$$r^2 - ms^2 = b.$$

This is equivalent to requiring that $r + s\sqrt{m} \in \mathbb{Z}[\sqrt{m}]$ has norm b. If m is negative then $\mathbb{Q}(\sqrt{m})$ is imaginary quadratic and by Dirichlet's unit theorem there are only finitely many elements of $\mathbb{Q}(\sqrt{m})$ which have this norm. Hence, when m is negative there are only finitely many values of r for which $\operatorname{sqf}(r^2-b) = m$.

If m is positive, let ϵ_m be the unique fundamental unit that exceeds 1 in the embedding of $\mathbb{Q}(\sqrt{m}) \hookrightarrow \mathbb{R}$ that sends $\sqrt{m} \mapsto \sqrt{m}$. Notice that after these choices $\epsilon_m > \frac{1}{2}(1+\sqrt{m})$. This is because, by standard properties of the fundamental unit, $\epsilon_m = a + b\sqrt{d}$ for $a, b > 0$ and also ϵ_m is in the ring of integers of $\mathbb{Q}(\sqrt{m})$.

If $r + s\sqrt{m} \in \mathbb{Z}[\sqrt{m}]$ has norm b then then any other element of $\mathbb{Q}(\sqrt{m})$ with this norm is, up to sign, a power of the fundamental unit ϵ_m times $r + s\sqrt{m}$. Suppose that α is an element (up to sign) of $\mathbb{Z}[\sqrt{m}]$ with norm b that has minimum Archimidean absolute value. Note that there is a unique minimum Archimedean absolute value because $\mathbb{Z}[\sqrt{m}]$ is discrete. Then, every element of $\mathbb{Z}[\sqrt{m}]$ with norm b is of form $\pm\alpha\epsilon_m^k$ for some $k \geq 0$. We know that $\epsilon_m > \frac{1}{2}(1+\sqrt{m}) > 1.2$. We want to show that the x-coordinate of $\alpha\epsilon_m^k$ grows exponentially in k, as this will imply that we cannot have too many values of r such that $\operatorname{sqf}(r^2-b) = m$. Writing $|\cdot|$ for the Archimedean absolute value, we see that if $r + s\sqrt{m} = \pm\alpha\epsilon_m^k$ then,

$$\begin{aligned} |\alpha\epsilon_m^k| &= |r + s\sqrt{m}| \\ &\leq |r| + |s\sqrt{m}| \\ &\leq |r| + \sqrt{r^2 - b} \\ &\leq 2|r|. \end{aligned}$$

Rearranging, this means that

$$|r| \geq \frac{\alpha\epsilon_m^k}{2}.$$

Because we require $|r| < C_\epsilon D^{1+\epsilon}$, we conclude that it is sufficient to look at

$$k < \frac{\log \frac{2C_\epsilon D^{1+\epsilon}}{\alpha}}{\log \epsilon_m} \leq \frac{(1+\epsilon)\log D + \log 2C_\epsilon}{\log \epsilon_m} \leq \frac{1+\epsilon}{\log 1.2}\log D + \frac{\log 2C_\epsilon}{\log 1.2} \leq C_{2,\epsilon}\log D$$

for some positive $C_{2,\epsilon}$. The point is that the number of possible r that yield any given value of $\operatorname{sqf}(r^2-b)$ is small, and thus the total number of possible r in this case is at most $C_{2,\epsilon}D^{1-2\delta}\log D$.

Putting the two subcases together, our total count is $C_{1,\epsilon}D^{\frac{1}{2}(1+\epsilon)+\delta} + C_{2,\epsilon}D^{1-2\delta}\log D$. Taking $\delta = \frac{1}{6}$, we can make our overall upper bound $(C_1 + C_2)D^{\frac{2}{3}+\epsilon} \ll_\epsilon D^{\frac{2}{3}+\epsilon}$.

(3) $\boldsymbol{f(x) = (x-a)(x^2 + b_1 x + b_2)}$: We will reduce to the previous case via a pair of coordinate transformations which map integral points to integral points and work for any twist of $y^2 = f(x)$. Start with the equation $dy^2 = f(x)$. Define $x' := 2^2 x$, $y' := 2^3 y$, $a' := 2^2 a$, $b'_1 := 2^2 b_1$, and $b'_2 := 2^4 b_2$. We have

$$dy'^2 = (x' - a')(x'^2 + b'_1 x' + b'_2),$$

which looks the same as before but now b'_1 is even. Thus, we may complete the square, taking $x'' := x' + {}^1\!/\!_2 b'_1$, $y'' := y'$, $a'' := a' + {}^1\!/\!_2 b'_1$, and $b'' := -b'_2 + {}^1\!/\!_4 b'_1$ to obtain

$$dy''^2 = (x'' - a'')(x''^2 - b'').$$

Overall, we have $x'' = 2^2x + 2b_1$, $y'' = 2^3y$ and hence integral points map to integral points. Because the transformation is independent of d, we may apply the logic of the previous case to $f(x'') = (x'' - a'')(x''^2 - b'')$ to obtain an upper bound on the number of twists of our original equation with integral points.

(4) **f is not monic**: Suppose $f(x) = \sum_{i=0}^{3} f_i x^i$. Consider $x' := f_3x$, $y' := f_3y$, and $f'(x) := x'^3 + f_x x'^2 + f_1 f_3 x' + f_0 f_3^2$. Then, $y'^2 = f'(x')$ and f' is monic. Furthermore, f' is reducible over $\mathbb{Q}$ because f is. Thus, by Gauss' Lemma f' is reducible over $\mathbb{Z}$ and we may pass to one of the previous cases as appropriate because integral points are mapped to integral points. □

4. The Asymptotics of $R(x)$ and $I(x)$ up to Constants: Theorem 4.1

Our main theorem in this section will be a version of Theorem 1.2 which is correct up to constants.

Theorem 4.1. *Assume that the abc-conjecture is true. Then,*

$$R(x) \asymp \frac{x}{\sqrt{\log x}} \quad \textit{and} \quad I(x) \asymp \frac{x}{\sqrt{\log x}}.$$

Because Clemm and Trebat-Leder proved the lower bound (Theorem 1.1), it is sufficient for us to prove the upper bound. Recall the definitions of $R_d(x)$ and $I_d(x)$ from Definition 2.11. We will prove Theorem 4.1 by first proving the following lemma.

Lemma 4.2. *Let d be good. Then,*

$$R_d(x) \ll \frac{(1+\epsilon(d, \frac{x}{d}))x}{|d|\sqrt{\log x}} \quad \textit{and} \quad I_d(x) \ll \frac{(1+\epsilon(d, \frac{x}{d}))x}{|d|\sqrt{\log x}},$$

where

$$\epsilon(d, \frac{x}{d}) \ll \sqrt{|d|} e^{-K \log^{1/2}(|\frac{x}{d}|)}$$

for some absolute constant K.

We will pass from Lemma 4.2 to Theorem 4.1 by summing $R_d(x)$ over all good d. By Theorem 1.5, the set of good d is very sparse and therefore the asymptotic dependence on x stays the same. Notice that Lemma 4.2 is unconditional, but we depend on the *abc*-conjecture in Theorem 4.1 only to perform the aforementioned aggregation. We keep track of the error in Lemma 4.2 because we ensure that its dependence on d is mild enough that summing over the set of good d does not cause problems.

Throughout this section, we will compute all our bounds while pretending that our quadratic fields $\mathbb{Q}(\sqrt{m})$ (in both the real and imaginary settings) are ordered by $|m|$ rather than by $\Delta_{\mathbb{Q}(\sqrt{m})}$. This will not change anything by more than a factor of 4 and thus is irrelevant to Lemma 4.2 and Theorem 4.1.

To prove Lemma 4.2, we will count those n which satisfy the first condition of Theorem 2.10. This means that we want to count n which are divisible only by primes q for which $\left(\frac{\epsilon_d d}{q}\right) = 1$.

Remark. Sieve theory provides a general method for counting the numbers up to x which are divisible only by a subset S of the primes. When S has positive natural density α, sieve theory gives us an upper bound of $\gg \frac{x}{\log^{1-\alpha} x}$ — this follows from summation by parts and standard results[4]. However, in our setting S is the set of primes satisfying the first condition of Theorem 2.10. By Chebotarev's density theorem applied to the multiquadratic extension $\mathbb{Q}(\sqrt{p_1}, \ldots, \sqrt{p_r})$, one can check that S has natural density $\frac{1}{2} + o_d(1)$ and our sieve-theoretic upper bound on $R_d(x)$ and $I_d(x)$ would end up being

$$\ll \frac{x}{d \log^{1/2+o_d(1)} x}.$$

[4]See for example Theorem A.1 of [4]

This is insufficient to prove Lemma 4.2. Furthermore, we would run into trouble when trying to sum these answers over all good d because without something like the generalized Riemann hypothesis we have no control over the dependence on d of the little-o term that arises from Chebotarev's density theorem. One can do better by exploiting the additional structure that the set S has. Indeed, this is the idea behind the theorem of Serre [17] (whose proof Serre attributes to Raikov, Wintner, and Delange) which Clemm and Trebat-Leder used in their lower bound[5]. By using Selberg-Delange theory (Theorem 2.2) — which generalized some of the ideas used in [17] — we can obtain the correct dependence on x, including the coefficient of $\frac{x}{\sqrt{\log x}}$, and bypass the issue of computing the density of S altogether.

Definition 4.3. We write $\chi_d(\cdot)$ to denote the quadratic Dirichlet character $\left(\frac{\epsilon_d d}{\cdot}\right)$.

Lemma 4.4. *The Dirichlet character χ_d is primitive with conductor*

$$m_d = \begin{cases} d & 2 \nmid d \\ 4d & 2 \mid d. \end{cases}$$

Proof. Suppose that $2 \nmid d$. Write $d = \pm p_1 \cdots p_r$. For any prime $q \neq 2$, we have by quadratic reciprocity that

$$\begin{aligned}
\chi_d(q) &= \left(\tfrac{\epsilon_d d}{q}\right) \\
&= \left(\tfrac{\epsilon_d \operatorname{sgn}(d)}{q}\right) \prod_{1\le i\le r} \left(\tfrac{q}{p_i}\right) \left(\tfrac{-1}{q}\right)^{\frac{p_i-1}{2}} \\
&= \left(\tfrac{\epsilon_d \operatorname{sgn}(d)}{q}\right) \left(\tfrac{-1}{q}\right)^{\sum_{1\le i\le r} \frac{p_i-1}{2}} \prod_{1\le i\le r} \left(\tfrac{q}{p_i}\right) \\
&= \left(\tfrac{\epsilon_d \operatorname{sgn}(d)}{q}\right) \left(\tfrac{\epsilon_d \operatorname{sgn}(d)}{q}\right) \prod_{1\le i\le r} \left(\tfrac{q}{p_i}\right) \\
&= \prod_{1\le i\le r} \left(\tfrac{q}{p_i}\right).
\end{aligned}$$

We can extend this to all n by noting that $\chi_d(2) = 1$ that characters are multiplicative. We see by the Chinese remainder theorem that χ_d is primitive with period $|d|$ and hence $m_d = |d|$.

If $2 \mid d$ then our criterion is slightly more complicated. Write $d = 2d' = \pm 2p_1p_2\cdots p_r$. Again for a prime $q \neq 2$,

$$\begin{aligned}
\chi_d(q) &= \left(\tfrac{\epsilon_d \operatorname{sgn}(d)}{q}\right) \left(\tfrac{2}{q}\right) \prod_{1\le i\le r} \left(\tfrac{p_i}{q}\right) \\
&= \left(\tfrac{-\operatorname{sgn}(d)}{q}\right) \left(\tfrac{2}{q}\right) \prod_{1\le i\le r} \left(\tfrac{q}{p_i}\right) \left(\tfrac{-1}{q}\right)^{\frac{p_i-1}{2}} \\
&= \left(\tfrac{-\operatorname{sgn}(d)}{q}\right) \left(\tfrac{\operatorname{sgn}(d)\epsilon_{d'}}{q}\right) \chi_{11}(q) \prod_{1\le i\le r} \left(\tfrac{q}{p_i}\right) \\
\chi_d(q) &= \left(\tfrac{-\epsilon_{d'}}{q}\right) \chi_{\pm 1 \pmod 8}(q) \prod_{1\le i\le r} \left(\tfrac{q}{p_i}\right),
\end{aligned}$$

[5]Sieve-theoretic lower bounds which make much weaker assumptions on S exist [8,14] but would have issues similar to those of the sieve-theoretic upper bounds were we to use them for our application.

where χ_{11} is defined in Section 2.1. Then, In this case, χ_d is primitive with period $8|d'| = 4|d|$ and hence $m_d = |4d|$. □

Definition 4.5. We define $N_d(x)$ to be the set of integers between 1 and x which are divisible only by primes in the kernel of χ_d. We also define $N_d^{\text{sqf}}(x)$ to be the subset of $N_d(x)$ which is squarefree.

We will ultimately use $N_d(\frac{x}{d})$ to upper bound $R_d(x)$ and $I_d(x)$. Let

$$a_n := \begin{cases} 1 & n \text{ is divisible only by primes } q \text{ s.t. } \chi_d(q) = 1 \\ 0 & \text{otherwise.} \end{cases}$$

We will obtain Lemma 4.2 by deriving the asymptotic behavior of $N_d(y) = \sum_{n \le y} a_n$. We can access the latter by studying the Dirichlet series defined on $\mathrm{Re}(s) > 1$ by

$$F_d(s) := \sum_{n \ge 1} a_n n^{-s}.$$

We can then use Selberg-Delange theory (Theorem 2.2), which gives us a Tauberian theorem that allows us to pass from properties of $F_d(s)$ to the asymptotics of $N_d(y)$.

Lemma 4.6. *Let χ_d and $F_d(s)$ be as above. On the region* $\mathrm{Re}(s) > 1$,

$$F_d(s) = C_d(s)\zeta(s)^{1/2} L(s, \chi_d)^{1/2}. \tag{9}$$

for some $C(s)$ holomorphic on $\mathrm{Re}(s) > \frac{1}{2}$.

Proof. Observe that F_d has an Euler product

$$F_d(s) = \prod_q (1 - a_q q^{-s})^{-1}.$$

We can write this as

$$\begin{aligned} F(s) &= \prod_{q \nmid m_d} (1 - q^{-s})^{-1/2} (1 - \chi_d(q) q^{-s})^{-1/2} \prod_{\chi_d(q) = -1} (1 - q^{-2s})^{1/2} \\ &= \zeta(s)^{1/2} L(s, \chi_d)^{1/2} \prod_{q | m_d} (1 - q^{-s})^{1/2} \prod_{\chi_d(q) = -1} (1 - q^{-2s})^{1/2}, \end{aligned}$$

noting that $\chi_d(q) = -1$ if and only if $q \nmid m_d$ and $\chi_d(q) \neq 1$.

Thus, we may take

$$C_d(s) = \prod_{q | m_d} (1 - q^{-s})^{1/2} \prod_{q \colon \chi_d(q) = -1} (1 - q^{-2s})^{1/2}$$

to reach the factorization in the lemma statement. □

Corollary 4.7. *Let χ_d be as above. We have*

$$|N_d(y)| = \frac{(c_d + \epsilon(d, y))y}{\sqrt{\log y}}, \tag{10}$$

where

$$c_d = \frac{1}{\Gamma(\frac{1}{2})} L(\chi_d, 1)^{1/2} \prod_{q | m_d} \left(1 - \frac{1}{q}\right)^{1/2} \prod_{q : \chi_d(q) = -1} \left(1 - \frac{1}{q^2}\right)^{1/2}$$

and

$$\epsilon(d, y) \ll \sqrt{d} e^{-K\sqrt{\log y}}.$$

Proof. Recall the factorization of $F_d(s)$ that we got from Lemma 4.6. We want to apply Theorem 2.2 with $\rho = 1/2$ and

$$G(s) = L(s, \chi_d)C_d(s), \tag{11}$$

since we know that $L(s, \chi_d)$ can be analytically continued to $\mathbb{C}$ and C_d is holomorphic for $\mathrm{Re}(s) > 1/2$. Take β in Theorem 2.2 to be any constant less than $1/2$. On the region $\mathrm{Re}(s) \geq 1 - \beta$, Theorem 2.3 tells us that $L(s, \chi_d) \ll d^{3/8+\epsilon})$ for any $\epsilon > 0$. We also have $|C_d(s)| \leq |\zeta(3/2)^{1/2}| \ll 1$. This means that Theorem 2.2 can be applied with $M = \sqrt{d}$, giving us (10) with constant

$$c_d = \frac{G(1)}{\Gamma(1/2)} + \epsilon(d, y)$$

and error

$$\epsilon(d, y) \ll \sqrt{d}e^{-K\sqrt{\log y}}.$$

□

Proof of Lemma 4.2. Recall that $R_d(x)$ and $I_d(x)$ are the number of n of the appropriate sign which satisfy all of the criteria in Theorem 2.10. Because the lower bound is already known, all we need is an upper bound. It suffices to count those n which satisfy just the first criterion of Theorem 2.10. This criterion is independent of the sign of n, so the same approach will work for $R_d(x)$ and $I_d(x)$.

Putting this together, we see that $R_d(x) \leq N_d(\frac{x}{|d|})$ because we require that $|dn| \leq x$. Plugging this into Corollary 4.7, we have

$$R_d(x) \leq \frac{(c_{|d|} + \epsilon(|d|, \frac{x}{|d|}))x}{|d|\sqrt{\log x - \log |d|}}.$$

We can replace $\sqrt{\log x - \log |d|}$ in the denominator with $\sqrt{\log x}$ without changing the form of the error term. Therefore,

$$R_d(x) \leq \frac{(c_|d| + \epsilon'(|d|, \frac{x}{|d|}))x}{|d|\sqrt{\log x}}$$

as desired. □

Proof of Theorem 4.1. We will give the proof for $R(x)$, but the same proof works for $I(x)$. If $\mathbb{Q}(\sqrt{m})$ admits a curve with good reduction everywhere then some good d must divide m. There could be multiple good d that divide a given m, but at least as an upper bound we have

$$R(x) \leq \sum_{\substack{d \text{ good} \\ |d| \leq x}} R_d(x)$$

Per Lemma 4.2, we have for any $y \leq x$ that[6]

$$\sum_{\substack{d \text{ good} \\ |d| \leq x}} R_d(x) \ll x \left(\sum_{\substack{d \text{ good} \\ |d| \leq y}} \frac{1 + \epsilon(d, \frac{x}{|d|})}{|d|\sqrt{\log x}} + \sum_{|d| \geq y} \frac{1}{|d|} \right), \tag{12}$$

We first want to show that the second term on the right-hand-side of (12) approaches 0 as y approaches infinity. By Lemma 2.1,

$$\sum_{|d| \geq y} \frac{1}{|d|} \ll y^{-1/3}.$$

Thus, if we take $y := \log^{3/2+\delta}(x)$ for $\delta > 0$ the second term on the right-hand-side of (12) will go to zero as x goes to infinity. In the first term of (12), the ratio $\frac{x}{d}$ becomes $(1 - o(1))x$ — note that

[6]To be clear, this is not the same y that was used in the statement of Corollary 4.7.

this $o(1)$ can be chosen independently of d because in the first term we've restricted to the case of $d \leq y$. By Lemma 4.2, the error term in this regime is

$$\epsilon(d,x) \ll \sqrt{|d|}e^{-K\log^{1/2}(\frac{x}{|d|})} = o(1)$$

again independently of d. Putting everything together, we can rewrite (12) as

$$\sum_{\substack{d \text{ good} \\ |d|\leq x}} \ll \frac{(1+o(1))x}{\sqrt{\log x}} \sum_{\substack{d \text{ good} \\ |d|\leq y}} \frac{1}{|d|}. \tag{13}$$

By Theorem 1.5 and Lemma 2.1, we know that the last sum is a constant, and hence

$$R(x) \ll \frac{x}{\sqrt{\log x}}. \tag{14}$$

as desired. □

5. Getting an Upper Bound on the Constant

We are now ready to begin the proof of our main theorem, which we recall for convenience.

Theorem 1.2.

$$R(x) \sim \frac{c_R x}{\sqrt{\log x}} \quad \textit{and} \quad I(x) \sim \frac{c_I x}{\sqrt{\log x}}$$

where

$$c_R = \sum_{\substack{d\in\mathbb{Z} \\ d \text{ good}}} \frac{c'_d c''_{d,R}}{d2^{\omega(d)}} \quad \textit{and} \quad c_I = \sum_{\substack{d\in\mathbb{Z} \\ d \text{ good}}} \frac{c'_d c''_{d,I}}{d2^{\omega(d)}},$$

where the set of good d is defined in Definition 2.8, c'_d is as in Corollary 5.3 and $c''_{d,R}$ and $c''_{d,I}$ are as in Lemma 5.4.

We will proceed in two steps. First, we will show that the expression in Theorem 1.2 is an upper bound on the correct constant. We will do so by proving a version of Lemma 4.2 with the correct constants and then repeating the argument we used to pass from Lemma 4.2 to Theorem 4.1. Recall that Lemma 4.2 followed from asymptotics for $|N_d(y)|$ that we proved in Corollary 4.7. Our first step will be to compute the asymptotics of $|N_d^{\text{sqf}}(y)|$, the size of the squarefree subset of $N_d(y)$. While we are at it, we will also compute the number of elements of $N_d^{\text{sqf}}(y)$ which are odd and even.

Lemma 5.1. *Let $F_d(s)$ be defined on $\mathrm{Re}(s) > 1$ by the Dirichlet series with coefficients $(a_n)_{n\geq 1}$ where*

$$a_n := \begin{cases} 1 & \textit{if } n \textit{ is squarefree and } \chi_d(q) = 1 \textit{ for every } q \mid n \\ 0 & \textit{otherwise.} \end{cases}$$

Then, $F_d(s)$ factors as

$$F_d(s) = C'_d(s)\zeta(s)^{1/2}L(s,\chi_d)^{1/2},$$

where

$$C'_d(s) = \zeta(2s)^{-1/2}\prod_{q\mid m_d}(1+q^{-s})^{1/2}\prod_{q:\chi_d(q)=1}(1-q^{-2s})^{1/2}$$

is holomorphic on $\mathrm{Re}(s) > \frac{1}{2}$.

Proof. Observe that we have the Euler product

$$\begin{aligned} F_d(s) &= \prod_q (1 + b_q q^{-s}) \\ &= \prod_q (1 - b_q q^{-s})^{-1} \prod_q (1 - b_q q^{-2s}). \end{aligned}$$

Per Lemma 4.6, we have

$$= C_d(s) L(s, \chi_d)^{1/2} \zeta(s)^{1/2} \prod_{q : \chi_d(q) = 1} (1 - q^{-2s}).$$

Then, we can take

$$\begin{aligned} C'_d(s) &= C_d(s) \prod_{q : \chi_d(q) = 1} (1 - q^{-2s}) \\ &= \zeta(2s)^{-1/2} \prod_{q | m_d} (1 + q^{-s})^{1/2} \prod_{q : \chi_d(q) = 1} (1 - q^{-2s})^{1/2} \end{aligned}$$

as in the lemma statement. □

Corollary 5.2. *For d odd and a_n as in Lemma 5.1, let*

$$a_{n,o} := \begin{cases} 1 & \text{if } a_n = 1 \text{ and } n \text{ is odd} \\ 0 & \text{otherwise.} \end{cases}$$

Then, the corresponding Dirichlet series with coefficients $(a_{n,o})_{n \geq 1}$ is

$$F_{d,o}(s) = (1 + 2^{-s}) F_d(s).$$

Proof. Consider the ratio between $C'_d(s) L(s, \chi_d)^{1/2}$ and what $C'_d(s) L(s, \chi_d)^{1/2}$ would be were $\chi_d(2) := -1$ (i.e. we forbid the prime 2). In $L(s, \chi_d)^{1/2}$, the factor of $(1 - 2^{-s})^{-1/2}$ would become a factor of $(1 + 2^{-s})^{-1/2}$. In $C'_d(s)$, we remove a factor of $(1 + 2^{-2s})^{1/2}$. Thus, our total change is

$$\frac{(1 + 2^{-s})^{-1/2}}{(1 - 2^{-s})^{-1/2} (1 + 2^{-2s})^{1/2}} = 1 + 2^{-s}$$

as desired. □

Corollary 5.3. *Let $N_d^{\text{sqf}}(y)$ denote the subset of $N_d(y)$ consisting of squarefree numbers, $N_d^{(\text{sqf},o)}(y)$ denote the subset of $N_d^{\text{sqf}}(y)$ consisting of odd numbers, and let $N_d^{(\text{sqf},e)}(y)$ denote the subset of $N_d^{\text{sqf}}(y)$ consisting of even numbers. Then,*

$$|N_d^{\text{sqf}}(y)| = \frac{(c'_d + \epsilon(d, y))}{\sqrt{\log x}},$$

for some

$$\epsilon(d, y) \ll \sqrt{d} e^{-K \sqrt{\log y}},$$

where

$$c'_d = \frac{1}{\Gamma(1/2) \zeta(2)^{1/2}} L(\chi_d, 1)^{1/2} \prod_{q | m_d} (1 + 1/q)^{1/2} \prod_{q : \chi_d(q) = 1} (1 - 1/q^2)^{1/2}.$$

If d is odd then

$$|N_d^{(\text{sqf},o)}(y)| = \frac{(2/3 c'_d + \epsilon(d, y))}{\sqrt{\log x}}$$

and

$$|N_d^{(\mathrm{sqf},e)}(y)| = \frac{(1/3c'_d + \epsilon(d,y))}{\sqrt{\log x}}.$$

Note that if d is even then $N_d(y)$ already consists only of odd numbers so in this case $N_d^{(\mathrm{sqf},o)}(y) = N_d^{\mathrm{sqf}}(y)$ and $N_d^{(\mathrm{sqf},e)}(y) = 0$.

Proof. The result for $|N_d^{\mathrm{sqf}}(y)|$ follows from Lemma 4.6 and Theorem 2.2, following the same logic as the proof of Corollary 4.7. The result for $N_d^{\mathrm{sqf},o}(y)$ then follows, noting that the added factor, evaluated at $s = 1$, is $\frac{3}{2}$. □

We are now ready to state the main lemma of this section, which gives a sharp version of Lemma 4.2.

Lemma 5.4. *Let d be good. Then,*

$$R_d(x) = \frac{(c'_d c''_{d,R} + \epsilon(d, \frac{x}{d}))x}{d2^{\omega(d)}\sqrt{\log x}} \quad \textit{and} \quad I_d(x) = \frac{(c'_d c''_{d,I} + \epsilon(d, \frac{x}{d}))x}{d2^{\omega(d)}\sqrt{\log x}},$$

where c'_d is as defined in Corollary 5.3,

$$c'_{R,d} := \begin{cases} \frac{11}{12} & d \equiv \pm 1 \pmod 8 \\ \frac{2}{3} & d \equiv \pm 3 \pmod 8 \quad \textit{and} \quad d > 0 \\ \frac{1}{6} & d \equiv \pm 3 \pmod 8 \quad \textit{and} \quad d < 0 \\ \frac{1}{8} & 2 \pmod 8 \quad \textit{and} \quad d > 0 \\ \frac{1}{8} & 6 \pmod 8 \\ 0 & \textit{otherwise,} \end{cases}$$

$$c'_{I,d} := \begin{cases} \frac{11}{12} & d \equiv 1 \pmod 8 \quad \textit{and} \quad d > 0 \\ \frac{11}{12} & d \equiv 7 \pmod 8 \quad \textit{and} \quad d < 0 \\ \frac{1}{6} & d \equiv 3 \pmod 8 \quad \textit{and} \quad d < 0 \\ \frac{2}{3} & d \equiv 5 \pmod 8 \quad \textit{and} \quad d > 0 \\ \frac{1}{8} & d \equiv \pm 2 \pmod 8 \quad \textit{and} \quad d < 0 \\ 0 & \textit{otherwise,} \end{cases}$$

and

$$\epsilon\left(d, \frac{x}{d}\right) \ll \sqrt{|d|}e^{-K\log^{1/2}(|\frac{x}{d}|)}.$$

The bulk of our work in this section lies in proving Lemma 5.4. Recall that in the proof of Lemma 4.2 it was sufficient to consider the first condition in Theorem 2.10. Now, we will need to consider all five conditions. Observe that while the first condition imposed a condition on the primes that were allowed to divide n, the next three impose conditions on the value of n modulo the primes dividing d and modulo 4 or 8. These thus correspond to (c) in Section 1.

To get a sense of why one might expect constants like those in Lemma 5.4 and to motivate our proof, it will help to consider a simple example.

5.1. **An Example: "Almost" Computing $I_{-1729}(x)$.** We will compute a variant of $I_d(x)$ for $d = -1729$. Note that d is the squarefree part of $(-1)^3 - 1728$, and -1 is coprime to 2 and 3 so -1729 is indeed good by Definition 2.8. Furthermore, because $-1729 \equiv -1 \pmod 8$ and we are interested in $I_d(x)$, we do not need to consider the last three conditions of Theorem 2.10.

Proposition 5.5. *Let $I'_{\text{-}1729}(x)$ be the number of positive $n \leq x$ such that $\mathbb{Q}(\sqrt{-1729n})$ admits an elliptic curve with good reduction everywhere and rational j-invariant. Then,*

$$I'_{\text{-}1729}(x) \sim \frac{c_d^{\text{sqf}} x}{4d\sqrt{\log x}}$$

where c'_d is as defined in Corollary 5.3.

The difference between $I_d(x)$ and $I'_d(x)$ is that the former orders the fields by discriminant and the latter orders by n.

Proof. Since we want imaginary fields, Theorem 2.10 says that $\mathbb{Q}(\sqrt{-1729n})$ admits an elliptic curve with good reduction everywhere and rational j-invariant when n is a positive number satisfying the following:

(a) n is coprime to 1729;
(b) n is squarefree;
(c) $\left(\frac{1729}{q}\right) = 1$ for every prime $q \mid n$;
(d) $\left(\frac{n}{7}\right) = \left(\frac{n}{13}\right) = \left(\frac{n}{19}\right) = 1. = \left(\frac{n}{13}\right) = \left(\frac{n}{19}\right) = 1.$

Observe that (c), by reciprocity (and as computed in the proof of Lemma 4.4), is equivalent to

$$\left(\tfrac{q}{7}\right)\left(\tfrac{q}{13}\right)\left(\tfrac{q}{19}\right) = 1 \text{ for every odd prime } q \mid n. \tag{15}$$

Let $(a_n)_{n\geq 1}$ be the sequence such that

$$a_n = \begin{cases} 1 & \text{if } n \text{ satisfies (a)-(c)} \\ 0 & \text{otherwise.} \end{cases}$$

Let $(b_n)_{n\geq 1}$ be any sequence such that

$$b_n = \begin{cases} 1 & \text{if } \left(\frac{n}{7}\right) = \left(\frac{n}{13}\right) = 1, \\ 0 & \text{if } \left(\frac{n}{7}\right)\left(\frac{n}{13}\right) = -1, \\ \text{arbitrary} & \text{if } (n, 91) \neq 1. \end{cases}$$

Observe that

$$I_{\text{-}1729}(x) = \sum_{n\leq x} a_n b_n,$$

since if we know that if $a_n = 1$ and $b_n = 1$ then by (15) and the definition of $(b_n)_{n\geq 1}$ we have $\left(\frac{n}{19}\right) = 1$ so (d) holds. Similarly, if $(n, 1729) \neq 1$ then $a_n = 0$ and it doesn't matter what b_n is. Therefore, $a_n b_n = 1$ if and only if (a)-(d) are satisfied.

By Lemma 5.1, we have that

$$\sum_{n\geq 1} a_n n^{-s} = C_d^{\text{sqf}}(s)\zeta(s)^{1/2} L(s, \chi_d)^{1/2},$$

where $C_d^{\text{sqf}}(s)$ is holomorphic on $\text{Re}(s) > \frac{1}{2}$. Observe that we may take $(b_n)_{n\geq 1}$ to be defined by

$$\sum_{n\geq 1} b_n n^{-s} = \left(\frac{\zeta(s) + L(s, (\frac{\cdot}{7}))}{2}\right) \otimes \left(\frac{\zeta(s) + L(s, (\frac{\cdot}{13}))}{2}\right)$$

where $\otimes$ denotes Rankin-Selberg convolution. To see this, notice that the first term on the right-hand-side is a Dirichlet series whose n^{th} term has coefficient 1 if n is a quadratic residue modulo 7, coefficient 0 if n is a nonresidue modulo 7, and coefficient $\frac{1}{2}$ if n is divisible by 7. The same is true for the second term on the right-hand-side and 13. The coefficient of the nth term of their convolution is thus 1 when n is a residue modulo 91, 0 if n is a nonresidue modulo either 7 or 13, and $\frac{1}{4}$ if n is divisible by 91. We are interested in

$$\sum_{n\geq 1} a_n b_n n^{-s} = \left(\sum_{n\geq 1} a_n n^{-s}\right) \otimes \left(\sum_{n\geq 1} b_n n^{-s}\right).$$

By the above, this is equal to

$$C_d(s)\zeta(s)^{1/2} L\left(s, \left(\tfrac{\cdot}{7}\right)\left(\tfrac{\cdot}{13}\right)\left(\tfrac{\cdot}{19}\right)\right)^{1/2} \otimes \left(\frac{\zeta(s) + L(s, \left(\tfrac{\cdot}{7}\right))}{2}\right) \otimes \left(\frac{\zeta(s) + L(s, \left(\tfrac{\cdot}{13}\right))}{2}\right),$$

and after distributing over the second convolution we have

$$C_d\left(s\right)\zeta\left(s\right)^{1/2} L\left(s, \left(\tfrac{\cdot}{7}\right)\left(\tfrac{\cdot}{13}\right)\left(\tfrac{\cdot}{19}\right)\right)^{1/2} \otimes \frac{1}{4}\left(\zeta\left(s\right) + L\left(s, \left(\tfrac{\cdot}{7}\right)\right) + L\left(s, \left(\tfrac{\cdot}{13}\right)\right) + L\left(s, \left(\tfrac{\cdot}{7}\right)\left(\tfrac{\cdot}{13}\right)\right)\right). \quad (16)$$

We want to distribute over the first convolution and then apply Theorem 2.2 to each term. Observe that for any Dirichlet character χ, we have

$$\left(C_d(s)\zeta(s)^{1/2} L(s, \chi_d)^{1/2}\right) \otimes L(s, \chi) = C_d'(s) L(s, \chi)^{1/2} L(s, \chi_d\chi)^{1/2} \quad (17)$$

where $C_d'(s)$ is absolutely convergent and hence holomorphic for $\mathrm{Re}(s) > \frac{1}{2}$. If neither χ nor $\chi_d\chi$ are nonprincipal we see that (17) has no singularity at $s = 1$. As long as one of the terms in (16) (each of which has the form in (17)) has a singularity at $s = 1$, the terms without singularities do not contribute to $\sum_{n\leq x} a_n b_n$ in the limit. Because χ_d is primitive with modulus 1729 and the characters to the right of the convolution in (16) have modulus strictly smaller than 1729, the only the only term of (16) with a nonprincipal character is the one corresponding to $\frac{1}{4}\zeta(s)$. Plugging this into (17), this is

$$\frac{1}{4} C_d(s)\zeta(s)^{1/2} L\left(s, \left(\tfrac{\cdot}{7}\right)\left(\tfrac{\cdot}{13}\right)\left(\tfrac{\cdot}{19}\right)\right)^{\frac{1}{2}}$$

since convolving any L-series with $\zeta(s)$ doesn't change anything. Therefore, we see that

$$I_{\text{-}1729}(x) = \sum_{n\leq x} a_n b_n \sim \frac{c_d x}{4\sqrt{\log x}}$$

where c_d is the constant from Corollary 5.3. □

The idea of this proof is that we don't need $(b_n)_{n\geq 1}$ to encode all of the information from the second constraint of Theorem 2.10 — it is sufficient to enforce that n is a residue modulo all but one of the primes dividing d as the first constraint of Theorem 2.10, encoded in $(a_n)_{n\geq 1}$, forces the correct behavior at the last prime.

Remark. We could have also included the factor associated with the last prime in $(b_n)_{n\geq 1}$. This would have introduced another main term after convolution because $L(s, \chi_d)$ would now appear in the expansion of $\sum_n b_n n^{-s}$, but the computation goes through in much the same way. In this case, each main term contributes $\frac{1}{2^{\omega(d)}}$ to the final constant.

5.2. **Proving Lemma 5.4.** The proof of the general case will proceed in a similar fashion to the proof of Proposition 5.5.

Proof. We will give the proof for $R_d(x)$, as the proof for $I_d(x)$ is analogous. We branch into three cases based on the conditions from Theorem 2.10 which are relevant to each: $d \equiv \pm 1 \pmod 8$, $d \equiv \pm 3 \pmod 8$, and d even. As before, if d is odd we write $d = \pm p_1 p_2 \ldots p_r$ and if d is even we write $d = 2d' = \pm 2p_1 p_2 \ldots p_r$. Abusing notation slightly, we will count positive numbers n such that $\mathbb{Q}(\sqrt{dn\,\mathrm{sgn}(d)})$ has discriminant with absolute value at most x and admits an elliptic curve with good reduction everywhere and rational j-invariant. Throughout this proof, we will write $o_d'(1)$ to denote an error bound which is $\ll \sqrt{|d|} e^{-K\sqrt{\log(\frac{x}{|d|})}}$ for some absolute constant K.

(1) $\boldsymbol{d \equiv \pm 1 \pmod 8}$: In this case, analogously to Proposition 5.5, we have the following constraints:
   (a) n is coprime to d;
   (b) n is squarefree;
   (c) $\chi_d(q) = 1$ for every $q \mid n$;
   (d) $\left(\frac{n}{p}\right) = \left(\frac{-\operatorname{sgn}(d)\epsilon_d}{p}\right)$ for every $p \mid d$.

Since d is odd, we have from Lemma 4.4 that

$$\chi_d(n) = \prod_{p \mid d} \left(\frac{n}{p}\right) \quad \text{and} \quad \chi_d(2n) = \chi_d(n)$$

when n is odd. There would be no hope of satisfying both (c) and (d) if

$$\prod_{p \mid d} \left(\frac{-\operatorname{sgn}(d)\epsilon_d}{p}\right) = -1.$$

This happens only if $\operatorname{sgn}(d)\epsilon_d = 1$ and an odd number of the p are congruent to 3 (mod 4). The latter condition is equivalent to $\operatorname{sgn}(d)\epsilon_d = -1$. Thus, this is never an issue[7].

Otherwise, we may assume that

$$\prod_{p \mid d} \left(\frac{-\operatorname{sgn}(d)\epsilon_d}{p}\right) = 1$$

Because we are interested in the ordering the quadratic fields $\mathbb{Q}(\sqrt{m})$ by discriminant rather than simply by m, we must also keep track of the congruence class of n modulo 4 so that when $m = dn \operatorname{sgn}(d) \equiv 2, 3 \pmod 4$ we only take values of n up to $\frac{x}{4d}$. We want to treat separately the cases where $nd \operatorname{sgn} d \equiv 1 \pmod 4$, $nd \operatorname{sgn} d \equiv 3 \pmod 4$, and n even. We denote by $R_d(x)\big|_{a(m)}$ the number of n such that $dn \operatorname{sgn}(d) \equiv a \pmod m$ and $\mathbb{Q}(\sqrt{dn \operatorname{sgn}(d)})$ has discriminant at most x. Let us start with $R_d(x)\big|_{1(4)}$, meaning that we want to add the conditions that
   (e) $\left(\frac{n}{4}\right) = \left(\frac{d \operatorname{sgn} d}{4}\right)$.

Let $(a_n)_{n \geq 1}$ be the sequence where $a_n = 1$ whenever n is odd and satisfies (a)-(c) and $a_n = 0$ otherwise. Let $(b_n)_{n \geq 1}$ be the sequence of coefficients of

$$\left(1/2 \left(\zeta(s) + \left(\frac{d \operatorname{sgn}(d)}{4}\right) L\left(s, \left(\frac{\cdot}{4}\right)\right)\right)\right) \otimes \left(\bigotimes_{i=1}^{r-1} 1/2 \left(\zeta(s) + \left(\frac{-\operatorname{sgn}(d)\epsilon_d}{p_i}\right)\right) L\left(s, \left(\frac{\cdot}{p_i}\right)\right)\right). \tag{18}$$

The term outside the big convolution corresponds to a Dirichlet series with a sequence of coefficients whose n^{th} term — assuming n is coprime to 4 — is 1 if $\left(\frac{n}{4}\right) = \left(\frac{d \operatorname{sgn}(d)}{4}\right)$ and 0 otherwise. Because $(a_n)_{n \geq 1}$ is 0 whenever n is even, we do not need to worry about the behavior of $(b_n)_{n \geq 1}$ when n is even.

Consider any other term of this convolution. It corresponds to a Dirichlet series with a sequence of coefficients whose n^{th} term — assuming n is coprime to a — is 1 if $\left(\frac{n}{p}\right) = \left(\frac{-\operatorname{sgn}(d)\epsilon_d}{p}\right)$ and 0 otherwise. Because $(a_n)_{n \geq 1}$ is 0 whenever n and d are not coprime by (a), we do not need to worry about the behavior of $(b_n)_{n \geq 1}$ when n and d are not coprime.

[7] In the imaginary case, this "compatibility" between (c) and (d) does matter. Condition (d) forces that

$$\left(\frac{n}{p}\right) = \left(\frac{\operatorname{sgn}(d)\epsilon_d}{p}\right)$$

for every $p \mid d$. In this case, there's no way that (c) can also be satisfied if $\operatorname{sgn}(d)\epsilon_d = -1$. As an aside, notice that this is exactly the fifth constraint of Theorem 2.10! This means that the fifth constraint of Theorem 2.10 is actually redundant for odd d.

Assuming that n satisfies (a), (b), and (c), we see that that $b_n = 1$ if and only if (e) is satisfied (because of the first term in (18)) and (d) is satisfied for all but one of the primes dividing d. Since

$$\prod_{p|d} \left(\tfrac{-\operatorname{sgn}(d)\epsilon_d}{p}\right) = 1$$

we see that knowing that all but one of the values of $\left(\frac{n}{p}\right)$ for $p \mid d$ tells us the last one. Thus, $a_n = b_n = 1$ if and only if (a)-(e) are satisfied. Because we have forced $n \equiv d \operatorname{sgn} d \pmod 4$ we have that

$$R_d(x)\big|_{1(4)} \equiv \sum_{n \le x} a_n b_n.$$

We will access this by applying Theorem 2.2 to

$$\sum_{n\ge 1} a_n b_n n^{-s} = \left(\sum_{n\ge 1} a_n n^{-s}\right) \otimes \left(\sum_{n\ge 1} b_n n^{-s}\right).$$

We know from Lemma 5.1 that

$$\sum_{n\ge 1} a_n n^{-s} = \zeta(s)^{1/2} L(s,\chi_d)^{1/2} C_d^{(\mathrm{sqf},o)}(s) \tag{19}$$

where $C_d^{(\mathrm{sqf},o)}(s)$ is absolutely convergent on $\operatorname{Re}(s) > 1/2$ — convolution does not change this property because $|\chi(n)| = 1$ for all n — and $c^{(\mathrm{sqf},o)} = L(1,\chi_d) C_d^{\mathrm{sqf}}(1)$. Expanding out the convolutions of $\sum_n b_n n^{-s}$ in (18) yields a sum of Dirichlet L-series, each associated with a product of Legendre symbols and the symbol checking for squares modulo 4. All of these characters are primitive and have modulus not equal to d because they are missing the last prime factor. Convolving our factorization of $\sum_{n\ge 1} a_n n^{-s}$ in (19) with any $L(s,\chi)$ will yield a term of form

$$L(s,\chi)^{1/2} L(s,\chi\chi_d)^{1/2} C'(s)$$

where $C'(s)$ is absolutely convergent on $\operatorname{Re}(s) > 1/2$. This term will not have a singularity at $s = 1$ unless at least one of χ or $\chi\chi_d$ is principal. Because χ_d is primitive with modulus d, this can only happen when χ is the trivial character. It thus suffices to apply Theorem 2.2 to the term

$$\left(\frac{1}{2^{\omega(d)}}\zeta(s)\right) \otimes \left(\zeta(s)^{1/2} L(s,\chi_d)^{1/2} C_d^{(\mathrm{sqf},o)}(s)\right) = \frac{1}{2^{\omega(d)}}\zeta(s)^{1/2} L(s,\chi_d)^{1/2} C_d^{(\mathrm{sqf},o)}(s),$$

which yields

$$R_d(x)\big|_{1(4)} = \sum_{n\le x} a_n b_n = \frac{(c_d^{(\mathrm{sqf},o)} + o_d'(1))}{|d| 2^{\omega(d)} \sqrt{\log x}}$$

where the sum is up to x because in this case $nd \operatorname{sgn}(d) \equiv 1 \pmod 4$.

We can run a similar argument for $R_d(x)|_{3(4)}$. In this case, we can take $\sum_{n\ge 1} b_n n^{-s}$ to be

$$\left(\frac{1}{2}\left(\zeta(s) - \left(\tfrac{d\operatorname{sgn}(d)}{4}\right) L\left(s, \left(\tfrac{\cdot}{4}\right)\right)\right)\right) \otimes \left(\bigotimes_{i=1}^{r-1} 1/2 \left(\zeta(s) + \left(\tfrac{-\operatorname{sgn}(d)\epsilon_d}{p_i}\right)\right) L\left(s, \left(\tfrac{\cdot}{p_i}\right)\right)\right),$$

but since the only surviving term corresponds to $\zeta(s)$ the sign change in the first term of the product is irrelevant. The same argument as above yields

$$R_d(x)\big|_{3(4)} \sum_{n\le \frac{x}{4}} a_n b_n = \frac{(c_d^{(\mathrm{sqf},o)} + o_d'(1))x}{4|d| \cdot 2^{\omega(d)} \sqrt{\log x}}.$$

When computing $R_d(x)\big|_{2(4)}$ we have the added conditions that

(e) n is even.

(f) $\left(\frac{n}{4}\right) = -\left(\frac{\operatorname{sgn} d}{4}\right)$.

This gives us

$$R_d(x)\big|_{2(4)} = \frac{(c_d^{(\mathrm{sqf},e)} + o'_d(1))x}{4|d| \cdot 2^{\omega(d)}\sqrt{\log x}}.$$

Adding all the contributions together and noting by Corollary 5.3

$$\frac{5}{4}c_d^{\mathrm{sqf},o} + \frac{1}{4}c_d^{\mathrm{sqf},e} = \frac{11}{12}c_d^{\mathrm{sqf}},$$

we see that when $d \equiv \pm 1 \pmod 8$,

$$R_d(x) = \frac{(\frac{11}{12}c_{\mathrm{sqf}}(d) + o'_d(1))}{|d|2^{\omega(d)}\sqrt{\log x}}$$

(2) $\boldsymbol{d \equiv \pm 3 \pmod 8}$: Conditions (a)-(d) still apply. In addition, we require that

(e) $n \equiv d \pmod 4$.

This means that for any given d we need to look at just one of the subcases in analysis for $d \equiv \pm 1 \pmod 8$. The one we need to look at is governed by $dn \operatorname{sgn}(d) \equiv \operatorname{sgn}(d) \pmod 4$. We thus have

$$R_d(x) = \frac{(a(d)c_d^{\mathrm{sqf},o} + o'_d(1))x}{|d|2^{\omega(d)}\sqrt{\log x}}$$

where

$$a(d) = \begin{cases} 1 & d > 0 \\ \frac{1}{4} & d < 0. \end{cases}$$

and replacing $c_d^{\mathrm{sqf},o}$ with $\frac{2}{3}c_d^{\mathrm{sqf}}$ yields the result.

(3) $\boldsymbol{d}$ **even**: As always, (a)-(d) from the analysis of the $d \equiv \pm 1 \pmod 8$ case still apply. We also have the additional condition that

(e) $n \equiv d + 1 \pmod 8$.

We will branch into subcases for $d \equiv 2 \pmod 8$ and $d \equiv 6 \pmod 8$. Note that these are the only two cases since d is squarefree. We will start with $d \equiv 6 \pmod 8$ as it is simpler.

When $d \equiv 6 \pmod 8$, we have from Lemma 4.4 that on odd n

$$\chi_d(n) = \chi_{11}(n) \prod_{1 \le i \le r} \left(\frac{n}{p_i}\right).$$

We only care about the behavior on odd n because (a) kills any even n. When (e) is satisfied $\chi_{11}(n) = 1$. We claim that

$$\prod_{i=1}^{r} \left(\frac{\operatorname{sgn} d}{p_i}\right) = 1 \tag{20}$$

and hence that (c)-(e) are consistent in this case. To see this, note that if $\operatorname{sgn} d = 1$ then (20) automatically holds. If $\operatorname{sgn} d = -1$, note that since $d \equiv 6 \pmod 8$ we have $d' \equiv 3 \pmod 4$ and hence an even number of the primes dividing d' are $3 \pmod 4$. Once again, we see that (20) holds.

As before, we now make the stronger claim that we only need to check (3), (5), and most of (4) to ensure that all three are satisfied. Suppose that $\chi_d(n) = 1$ and also $n \equiv 7 \pmod 8$ and for $1 \le i \le r - 1$,

$$\left(\frac{n}{p_i}\right) = \left(\frac{\operatorname{sgn} d}{p_i}\right).$$

Then, we see that for such n

$$\begin{aligned} 1 &= \chi_d(n) \\ &= \chi_{11} \prod_{1\le i\le r} \left(\tfrac{n}{p_i}\right) \\ &= \left(\tfrac{n}{p_r}\right) \prod_{i=1}^{r-1} \left(\tfrac{\operatorname{sgn} d}{p_i}\right) \end{aligned}$$

and comparing with (20) we have that

$$\left(\tfrac{n}{p_r}\right) = \left(\tfrac{\operatorname{sgn} d}{p_r}\right).$$

The point is that, as in the previous cases, it is sufficient to use $\sum_n b_n n^{-s}$ to check that $n \equiv 7 \pmod 8$ and the fourth condition for all but the last odd prime dividing d. We already know how to enforce the residuosity conditions. To force $n \equiv 7 \pmod 8$, consider the function $\mathbf{1}_{7(8)} \colon (\mathbb{Z}/8\mathbb{Z})^\times \to \mathbb{C}$ that is 1 on 7 (mod 8) and 0 everywhere else. The Fourier expansion of this function is

$$\mathbf{1}_{7(8)} = \tfrac{1}{4}(\chi_{00} - \chi_{01} - \chi_{10} + \chi_{11}).$$

The additional factor we will add to our expression for $\sum_n b_n n^{-s}$ in this case is thus

$$\tfrac{1}{4}(\zeta(s) - L(s,\chi_{01}) - L(s,\chi_{10}) + L(s,\chi_{11})),$$

where we can use $\zeta(s)$ instead of $L(s,\chi_{00})$ because we do not care about what happens when $(n,8) \neq 1$. Therefore, we may take $\sum_{n\ge1} b_n n^{-s}$ to be

$$\tfrac{1}{4}(\zeta(s) + L(s,\chi_{01}) - L(s,\chi_{10}) + L(s,\chi_{11})) \otimes \left(\bigotimes_{i=1}^{r-1} \tfrac{1}{2}\left(\zeta(s) + \left(\tfrac{\operatorname{sgn} d}{p_i}\right) L\left(s, \left(\tfrac{n}{p_i}\right)\right)\right)\right).$$

Choosing $(a_n)_{n\ge1}$ as in the previous steps and running the same logic as before, we see that the only term of $\sum_n b_n n^{-s}$ that yields a term with a singularity at $s=1$ after convolving with $\sum_n a_n n^{-s}$ is the term corresponding to $\zeta(s)$. Therefore,

$$R_d(x) = \sum_{n\le \frac{x}{4}} a_n b_n = \frac{(\frac{1}{8}c_d^{\mathrm{sqf}} + o_d'(1))}{|d| 2^{\omega(d)} \sqrt{\log x}}.$$

When $d \equiv 2 \pmod 8$ we need to be a little more careful. In this case,

$$\chi_d(n) = \chi_{01} \prod_{1\le i\le r} \left(\tfrac{n}{p_i}\right).$$

However, unlike when $d \equiv 6 \pmod 8$ the expression

$$\prod_{i=1}^{r} \left(\tfrac{\operatorname{sgn}(d)}{p_i}\right)$$

is -1 when $\operatorname{sgn}(d) < 0$. Thus, it is not possible to satisfy (c)-(e) simultaneously! If we assume $\operatorname{sgn}(d) > 0$ then the logic proceeds just as before and we get the same answer. $\square$

6. A Matching Lower Bound: Proving Theorem 1.6 and Theorem 1.2

Consider the upper bound on $R(x)$ (the same logic works for $I(x)$ as well). Using Lemma 5.4 instead of Lemma 4.2 in the proof of Theorem 4.1 tells us that

$$R(x) \leq \sum_{d \text{ good}} R_d(x) \sim \frac{x}{\sqrt{\log x}} \left(\sum_{d \text{ good}} \frac{c_d}{d} + o(1) \right). \tag{21}$$

This is a priori only an upper bound because we may be double-counting – if two good d_1 and d_2 both divide m then $\mathbb{Q}(\sqrt{m})$ may be double-counted in (21) since it could be that $\frac{m}{d_1}$ contributes to $R_{d_1}(x)$ and $\frac{m}{d_2}$ contributes to $R_{d_2}(x)$. Write $R_{d_1,d_2}(x)$ to be the set of positive numbers n such that $K = \mathbb{Q}(\sqrt{\operatorname{lcm}(d_1,d_2)n})$ admits an elliptic curve with good reduction everywhere and Δ_K is at most x. Then, by inclusion-exclusion, we have the lower bound

$$R(x) \geq \sum_{d \text{ good}} R_d(x) - \sum_{d,d' \text{ good}} R_{d,d'}(x). \tag{22}$$

The idea is that this second term ends up being $\asymp \frac{x}{\log^{3/4} x}$ and hence is negligible compared to the first term (which is $\asymp \frac{x}{\sqrt{\log x}}$ by Theorem 4.1). As in the proof of Theorem 4.1, there are two parts to this result. We will first show that the sum of $\frac{1}{\operatorname{lcm}(d,d')}$ over good values of d and d' is constant. Then, we will show that the dependence on x of any $R_{d,d'}(x)$ is $\asymp \frac{x}{\log^{3/4} x}$. For the former, we will show that the set of least-common multiples up to x of good d (counted with multiplicity) is $\ll x^{1-\kappa}$ for some $\kappa > 0$. Then, the first part of Lemma 2.1 will us that the sum of reciprocals of least-common-multiples with multiplicity is some absolute constant.

Theorem 1.6. *Given a set $S \subseteq \mathbb{N}$ of squarefree numbers and positive constants B and $\beta < 1$ such that*

$$\#\{n \leq x \colon n \in S\} \leq Bx^{1-\beta}$$

for all positive x, there are positive constants C and $\kappa < 1$ such that

$$\#\{(n,n') \colon n \in S, n' \in S, \operatorname{lcm}(n,n') \leq x\} \leq Kx^{1-\kappa}.$$

for all positive x.

The squarefree condition is not important but we include it because it makes the proof simpler and holds for our application, where S will be the set of good d. As motivation for Theorem 1.6, notice that the analogous result where least-common-multiple is replaced by product holds with $\kappa = \beta - o(1)$. To see this, consider splitting the range $[1,x]$ into intervals $(y,2y]$. For each $a \in (y,2y]$, any $b \in [1,x]$ such that $ab \leq x$ is at most $\frac{x}{y}$. There are at most $B\left(\frac{x}{y}\right)^{1-\beta}$ such values of b. The number of possible values of $a \in (y,2y]$ is at most $B(2y)^{1-\beta}$. Therefore, the number of tuples (a,b) where $a \in (y,2y]$ is at most

$$B^2 2^{1-\beta} y^{1-\beta} \left(\frac{x}{y}\right)^{1-\beta} = B^2 2^{1-\beta} x^{1-\beta}.$$

This is uniform in y and there are at most $\log x + 1$ intervals $(y,2y]$. Therefore, the total number of pairs (a,b) which work is at most $B^2 2^{1-\beta} x^{1-\beta}(\log x + 1) \ll x^{1-\beta+o(1)}$. The structure of our proof of Theorem 1.6 is similar. The main difficulty is that $\operatorname{lcm}(a,b)$ may be much smaller than ab so it is harder to control the number of b which can be associated to a given a.

Proof. We will count the number of tuples (a,b) where $a \in (y,2y]$, $b \in (z,2z]$, and $\operatorname{lcm}(a,b) \leq x$. If we show that this is $\ll x^{1-\kappa}$ for some $\kappa > 0$ then aggregating over the $\ll \log^2 x$ choices of y and z gives us the desired result. Fix some $a \in (y,2y]$. We say that $b \in (z,2z]$ "works" with

a if $\operatorname{lcm}(a,b) \le x$. Because $\operatorname{lcm}(a,b) = \frac{ab}{\gcd(a,b)}$, a given value of b works with a if and only if $\gcd(a,b) \ge \frac{ab}{x}$. Let us count the number of factors of a which exceed $\frac{ab}{x} = \frac{a}{\frac{x}{b}}$. These are the possible values of $\gcd(a,b)$. Large factors of a number are in bijection with its small factors, so this is equivalent to counting the number of factors that a has up to $\frac{x}{b} \le \frac{x}{z}$. We will now branch into two cases depending on the sizes of y and z. In what follows, let $\delta := \frac{\beta(1-\beta)}{(1+2\beta-\beta^2)}$ and $\kappa = \frac{2\beta^2}{1+2\beta-\beta^2} - \eta$ for any sufficiently small positive η.

(1) $\boldsymbol{\min(y,z) \ge x^{1-\delta}}$: There are at most $B(2y)^{1-\beta}$ choices for a. The number of factors of a which are at most $\frac{x}{z}$ is trivially at most $\frac{x}{z}$, and therefore the number of factors of a exceeding $\frac{ab}{x}$ is at most $\frac{x}{z}$. For any such factor, the number of possible b which also divide this factor is at most

$$\frac{z}{\frac{ab}{x}} = \frac{zx}{ab} \le \frac{x}{y}$$

since $b \in (z, 2z]$. Putting everything together, the number of tuples (a,b) from this pair of intervals is at most

$$B(2y)^{1-\beta} \cdot \frac{x}{z} \cdot \frac{x}{y} = \frac{2^{1-\beta}Bx^2}{zy^\beta} \le 2^{1-\beta}Bx^{2-(1-\delta)(1+\beta)} = 2^{1-\beta}Bx^{1+\delta-\beta+\delta\beta}.$$

We are satisfied as long as

$$1+\delta-\beta+\delta\beta \le 1-\kappa \iff \delta \le \frac{\beta-\kappa}{1+\beta},$$

and this holds for our choice of δ and κ.

(2) $\boldsymbol{\min(y,z) \le x^{1-\delta}}$: Without loss of generality, assume that $z \le y$. Observe that a has $\le C\frac{\log y}{\log\log y}$ prime factors (because a is squarefree) for some absolute constant C, and we are interested in knowing how many numbers up to $\frac{x}{z}$ are divisible only by these prime factors. Because we just want an upper bound, we can take the allowable prime factors (regardless of what a is) to be the prime numbers up to $D\log y$[8]. In the notation of Proposition 2.4, we thus are interested in $\Psi(\frac{x}{z}, C\log y)$. By assumption, we have $\frac{x}{z} > x^\delta$ and hence $C\log y \le C\log x \le \log^{1+\epsilon+o(1)}\frac{x}{z}$ for any $\epsilon \ge 0$. If we take ϵ to be some arbitrarily small nonzero value, applying Proposition 2.4 then tells us that

$$\Psi\left(\frac{x}{z}, C\log y\right) = \left(\frac{x}{z}\right)^{1-\frac{1}{1+\epsilon}+o(1)} = \left(\frac{x}{z}\right)^{\epsilon'+o(1)}$$

for some ϵ' which can be chosen to be arbitrarily small. Remember that this is an upper bound on the number of factors of a which exceed $\frac{ab}{x}$. Given a choice of a and such a factor, the number of possible choices for b is $\min(\frac{x}{y}, B(2z)^{1-\beta})$ where the former term arises by the same logic as was used in the previous case and the latter term arises as an upper bound on the size of $S \cap (z, 2z]$. Putting things together, we have that the number of tuples (a,b) from this pair of intervals is at most

$$(2y)^{1-\beta} \cdot \left(\frac{x}{z}\right)^{\epsilon'+o(1)} \cdot \min\left(\frac{x}{y}, B(2z)^{1-\beta}\right) \le 2^{1-\beta}\min\left(\frac{x^{1+\epsilon'+o(1)}}{y^\beta}, B(4yz)^{1-\beta}\right).$$

If $y > x^{\frac{\kappa+\nu}{\beta}}$ (for some $\nu > 0$ which we can take to be arbitrarily small) then

$$\frac{x^{1+o(1)}}{y^\beta} \le x^{1-\kappa-\nu+o(1)},$$

[8]Let T be a set of primes $p_1 < p_2 < \cdots < p_{|T|}$. Label a number $n = p_{i_1}p_{i_2}\dots p_{i_k}$ by the tuple $(i_1,\dots,i_k)$. The number n associated with every given tuple is minimized when we take T to be the first $|T|$ primes.

which is enough for x large enough. Otherwise, if $y < x^{\frac{\kappa+\nu}{\beta}}$, we have

$$B(4yz)^{1-\beta} \leq 4^{1-\beta} B x^{(1-\beta)(\frac{\kappa+\nu}{\beta}+1-\delta)} \leq 4^{1-\beta} B x^{1-(\kappa+\nu)+\frac{\kappa+\nu}{\beta}-\delta-\beta+\beta\delta}.$$

We are satisfied as long as

$$\frac{\kappa+\nu}{\beta} - \beta - \delta + \beta\delta < 0 \iff \frac{\kappa+\nu-\beta^2}{\beta(1-\beta)} < \delta,$$

and this again holds for our choices of δ and κ if ν is sufficiently small and x is sufficiently large. □

Applied to the set of good d, which by Theorem 1.5 satisfies the conditions of Theorem 1.6 for $\beta = \frac{1}{3}$, we can take $\kappa = \frac{1}{8}$.

Lemma 6.1.

$$R_{d,d'}(x) \ll \frac{(1+o'_{dd'}(1))x}{\operatorname{lcm}(|d|,|d'|)\log^{\frac{3}{4}} x} \quad \textit{and} \quad I_{d,d'}(x) \ll \frac{(1+o'_{dd'}(1))x}{\operatorname{lcm}(|d|,|d'|)\log^{\frac{3}{4}} x}.$$

Proof. As in the proof of Lemma 4.2, we will use only the first constraint from Theorem 2.10 and imagine that fields $\mathbb{Q}(\sqrt{m})$ are ordered by m rather than by discriminant. Let S (S') be the set of primes q such that $\chi_d(q) = 1$ ($\chi_{d'}(q) = 1$). An upper bound on $R_{d,d'}(x)$ is the number of n at most $\frac{x}{\operatorname{lcm}(d,d')}$ which are divisible only by primes in $S \cap S'$[9]. We have that when q is coprime to d and d',

$$q \in S \cap S' \implies \frac{1}{4}(\chi_d(q)+1)(\chi_{d'}(q)+1) = 1$$

$$q \notin S \cap S' \implies \frac{1}{4}(\chi_d(q)+1)(\chi_{d'}(q)+1) = 0.$$

The expression on the left-hand-side can be written as

$$\frac{1}{4}(1+\chi_d(q)+\chi_{d'}(q)+\chi_d\chi_{d'}(q).$$

Let

$$b_n := \begin{cases} 1 & n \text{ is divisible only by primes in } S \cap S' \\ 0 & \text{otherwise.} \end{cases}$$

Then, we have as in the proof of Lemma 4.2 that

$$F(s) := \sum_{n\geq 1} a_n n^{-s} = (\zeta(s)L(\chi_d,s)L(\chi_{d'},s)L(\chi_d\chi_{d'},s))^{\frac{1}{4}} C_{d,d'}(s),$$

where $C_{d,d'}(s)$ is holomorphic on $\operatorname{Re}(s) > \frac{1}{2}$ and has absolute value at most 1 regardless of the values of d and d'. Note that χ_d and $\chi_{d'}$ are primitive and nonprincipal, and they have different conductors when $d \neq d'$. Then, applying Theorem 2.2 with

$$F(s) = \zeta^{1/4}(s)G(s)$$

for

$$G(s) = (L(\chi_d,s)L(\chi_{d'},s)L(\chi_d\chi_{d'},s))^{1/4} C_{d,d'}(s)$$

gives us the result. □

Lemma 6.2.

$$\sum_{\substack{d,d' \text{ good} \\ d,d' \leq x}} R_{d,d'}(x) \ll \frac{x}{\log^{3/4} x}$$

[9]Note that this condition is necessary but not sufficient, as d and d' must themselves be compatible in the sense that the primes dividing $\frac{d'}{\operatorname{lcm}(d,d')}$ must lie in S and vice-versa. However, even this weak condition will be enough.

Proof. This follows from the same logic as was used in deriving Theorem 4.1 from Lemma 4.2. We have for any $y \leq x$ that

$$\sum_{\substack{d,d' \text{ good} \\ d,d' \leq x}} R_{d,d'}(x) \ll x \left(\sum_{\substack{d,d' \text{ good} \\ d,d' \leq y}} \frac{1 + o'_d(1)}{\log^{3/4} x} + \sum_{\substack{d,d' \text{ good} \\ d,d' > y}} \frac{1}{\operatorname{lcm}(d,d')} \right).$$

Taking $y := \log^{21/4+\delta} x$ for any $\delta > 0$, we see that the second term in the parenthesis goes 0 to as y goes to infinity by Theorem 1.5, Theorem 1.6 and Lemma 2.1. Every $o'_d(1)$ for $d \leq y$ is upper bounded by some $o(1)$. Therefore, as x goes to infinity we have a limit of

$$\frac{x}{\log^{3/4} x} \left(\sum_{d,d' \text{ good}} \frac{1}{\operatorname{lcm}(d,d')} \right) \ll \frac{x}{\log^{3/4} x}$$

as desired. □

Theorem 1.2 then follows from Lemma 5.4, (21) and (22), and Theorem 1.6.

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